

Project Data Page:

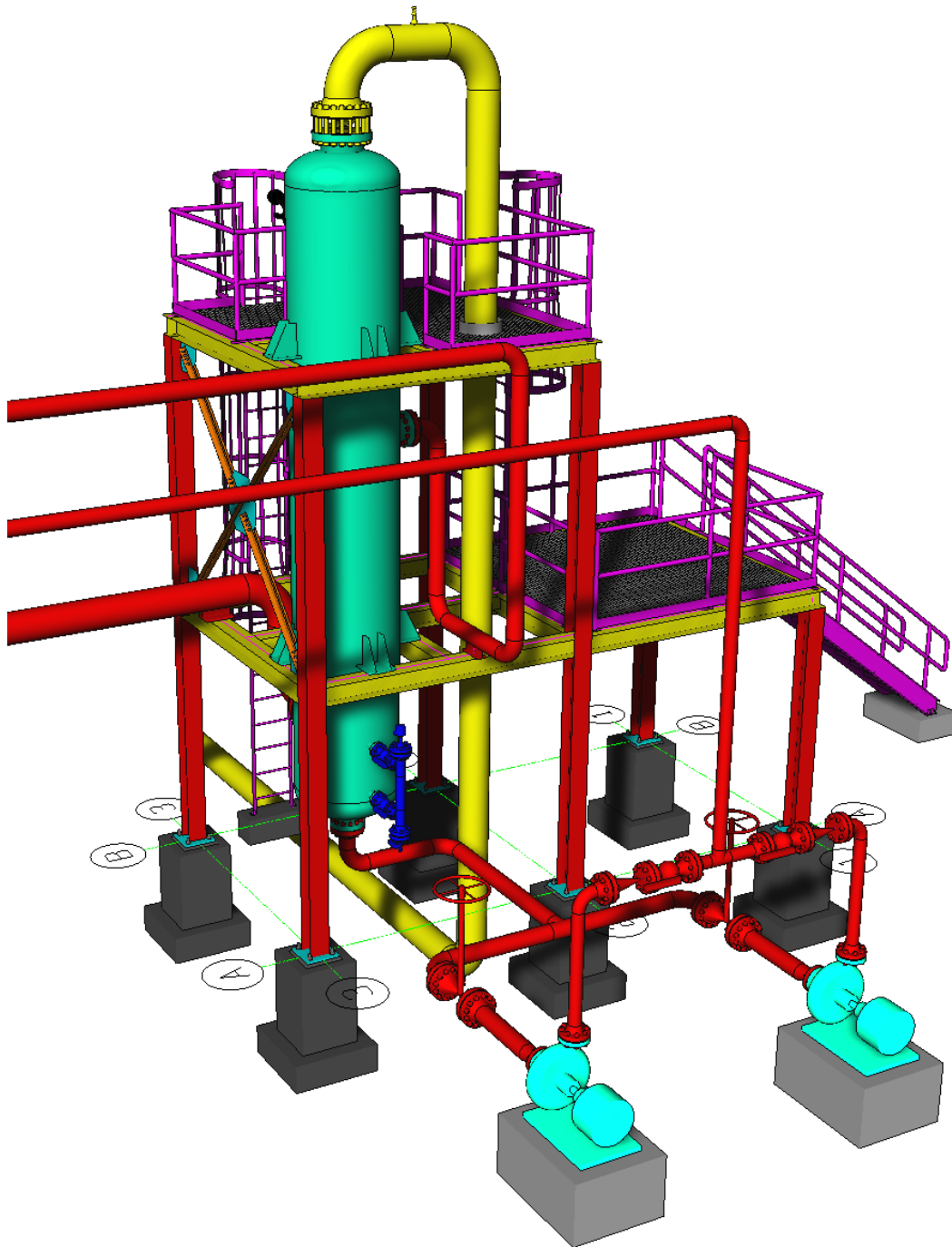




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CCW Expansion Vessel (V8804)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : D:\QUEP\CCW Expansion Vessel (V8804)-7%.pvdb

Date of Analysis : Sep 16, 2026 3:03pm

PV Elite 28, January 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 4000.000 mm. ID

Vessel Design Length, Tangent to Tangent 10500.00 mm.

Specified Datum Line Distance 0.00 mm.

Internal Design Temperature 82 °C

Internal Design Pressure 3.500 bars

External Design Temperature 82 °C

External Design Pressure 0.517 bars

Maximum Allowable Working Pressure 5 bars

External Max. Allowable Working Pressure 1 bars

Shop Test Pressure 4.550 bars

Field Test Pressure 4.550 bars

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
left Dish	3.859	0.52	5.000	1.6000	No	No
Shell-1	3.859	0.52	6.000	1.6000	N/A	No
Shell2	3.859	0.52	6.000	1.6000	N/A	No
Shell3	3.859	0.52	6.000	1.6000	N/A	No
shell4	3.859	0.52	6.000	1.6000	N/A	No
Right dish	3.859	0.52	5.000	1.6000	No	No

Liquid Level: 3700.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.989**Stiffener Ring Specifications:**

Elevation mm.	Selected Type	Ring Description
1313.00	Bar 175.0 x 12.0	Ring R1 Fr20
3938.00	Bar 175.0 x 12.0	Ring R1 Fr30
6560.00	Bar 150.0 x 12.0	Ring R1 Fr40



9285.00

Bar 150.0 x 12.0

Ring R1 Fr50

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	12.0	10.0	8.2	...	0.85	0.85
Cylinder	3050.0	3000.0	12.0	12.0	8.2	...	0.85	0.85
Cylinder	6050.0	3000.0	12.0	12.0	8.2	...	0.85	0.85
Cylinder	7950.0	1900.0	12.0	12.0	8.2	...	0.85	0.85
Cylinder	10450.0	2500.0	12.0	12.0	8.2	...	0.85	0.85
Ellipse	10500.0	50.0	12.0	10.0	8.2	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	10	8.96001	0.51711	0.67361
20	Ring	12	8.38473	0.51711	1.53293
Ring	30	12	9.82763	0.51711	0.93802
30	Ring	12	9.82763	0.51711	0.93802
Ring	40	12	9.82405	0.51711	0.93914
40	Ring	12	9.82405	0.51711	0.93914
Ring	50	12	9.94664	0.51711	0.90195
50	Ring	12	9.94664	0.51711	0.90195
Ring	70	12	8.2084	0.51711	1.63783
70	80	10	8.96001	0.51711	0.67361

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1646.33	4667.51	No Calc	No Calc
Ring	30	2625	4666.98	494.779	1489.17
30	Ring	2625	4666.98	No Calc	No Calc
Ring	40	2622	4666.98	606.765	1489.17
40	Ring	2622	4666.98	No Calc	No Calc
Ring	50	2725	4667.24	620.103	994.034
50	Ring	2725	4667.24	No Calc	No Calc
Ring	70	1548.33	4667.32	496.022	994.034
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.946	n/a	Passed
Lift Lug	WRC-107/537	0.946	n/a	Passed
N3-4"	WRC-297	0.885	Shell	Passed
N4-4"	WRC-297	0.885	Shell	Passed
N1-18	WRC-297	0.969	Shell	Passed
N2-8"	WRC-297	0.469	Nozzle	Passed
N5-4"	WRC-297	0.885	Shell	Passed
N6-4"	WRC-297	1.008	Shell	Failed

Foundation/Support Summary**Saddle Parameters:**

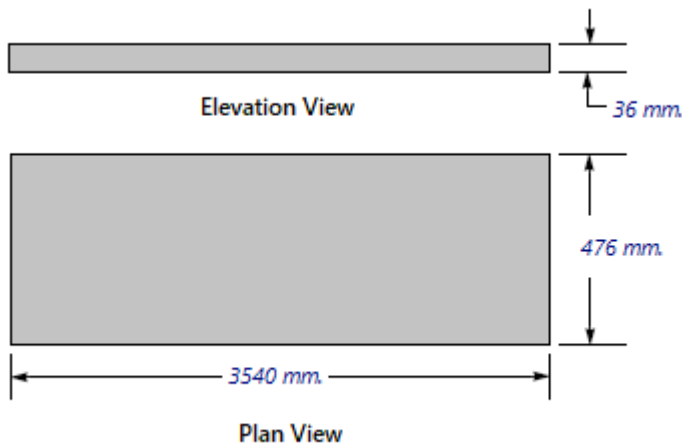
Saddle Width

250.000 mm.



Saddle Bearing Angle	150.000	deg.
Centerline Dimension	2260.000	mm.
Wear Pad Width	500.000	mm.
Wear Pad Thickness	12.000	mm.
Wear Pad Bearing Angle	162.000	deg.
Distance from Saddle to Tangent	2150.000	mm.
Baseplate Length	3540.000	mm.
Baseplate Thickness	36.000	mm.
Baseplate Width	476.000	mm.
Number of Ribs (including outside ribs)	5	
Rib Thickness	22.000	mm.
Web Thickness	22.000	mm.
Height of Center Web	200.000	mm.
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	100187.20	Kgf
Maximum Transverse Saddle Shear Load	5724.37	Kgf
Maximum Longitudinal Saddle Shear Load	11448.73	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	91217.57	Kgf
Maximum Transverse Saddle Shear Load	780.00	Kgf
Maximum Longitudinal Saddle Shear Load	959.75	Kgf

Weights:

Fabricated - Bare w/o removable internals	22741.4	kg.
Shop Test - Fabricated + water (full)	171379.3	kg.
Shipping - Fab. + rem. intls.+ shipping app.	22741.4	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	30692.2	kg.
Empty - Fab. + intls. + details + wghts.	30692.2	kg.
Operating - Empty + operating liquid (no ca)	173004.9	kg.
Field Test - Empty weight + test medium (full)	179330.2	kg.

**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1-24"	5	OK	...	...		OK	Passed
K2B-2"	6	OK	...	OK	6.40	OK	Passed
K2A-2"	6	OK	...	OK	6.40	OK	Passed
N3-4"	5	OK	...	OK	6.86	OK	Passed
N4-4"	5	OK	...	OK	6.86	OK	Passed
K3-2"	6	OK	...	OK	6.40	OK	Passed
K4-2"	6	OK	...	OK	6.40	OK	Passed
N1-18	6	OK	...	OK	8.76	OK	Passed
N2-8"	67	OK	...	OK	4.10	OK	Passed
N5-4"	5	OK	...	OK	6.86	OK	Passed
N6-4"	5	OK	...	OK	6.86	OK	Passed
K1A-2"	6	OK	...	OK	6.40	OK	Passed
K1B-2"	6	OK	...	OK	6.40	OK	Passed
M2-24"	5	OK	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5 Nozzle : M2-24"
 Minimum MAWP Shells/Flanges : 5 Element : Right dish
 Minimum MAPnc Shells/Flanges : 6 Element : Right dish

Computed Vessel M.A.W.P. : 5 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	M1-24"	0.000	0.000	1113.600
20	K2B-2"	650.000	180.000	108.000
20	K2A-2"	450.000	0.000	108.000
20	N3-4"	1050.000	0.000	187.794
20	N4-4"	2550.000	0.000	187.794
30	K3-2"	3450.000	0.000	108.000
30	K4-2"	4050.000	0.000	108.000
40	N1-18	7049.999	180.000	840.800
40	N2-8"	0.000	180.000	380.631
40	N5-4"	7249.999	0.000	187.794
40	N6-4"	7650.000	0.000	187.794
50	K1A-2"	9550.000	0.000	108.000
50	K1B-2"	9450.000	180.000	108.000
70	M2-24"	0.000	0.000	1113.600

The nozzle spacing is computed by the following:

$$= \sqrt{(l_l^2 + l_c^2)} \text{ where}$$

l_l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	218.82	300
K2A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.09	300
K3-2"	2.000 in	Actual	WNF	3.323	16.800	...	...	442.60	300
K4-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.09	300
K1A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	490.09	300
K1B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	218.82	300
N3-4"	4.000 in	160	WNF	4.500	13.487	...	...	426.96	300
N4-4"	4.000 in	160	WNF	4.500	13.487	...	...	481.07	300
N5-4"	4.000 in	160	WNF	4.500	13.487	...	...	426.96	300
N6-4"	4.000 in	160	WNF	4.500	13.487	...	...	426.96	300
N2-8"	8.000 in	120	WNF	8.625	18.263	...	...	251.07	150
N1-18	18.000 in	Actual	WNF	18.000	20.000	...	...	783.11	150
M1-24"	24.000 in	Actual	WNF	24.000	28.000	909.60	12.00	272.90	150
M2-24"	24.000 in	Actual	WNF	24.000	28.000	909.60	12.00	272.90	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2B-2"	SA-105	12.000	10.000	...	...	...
K2A-2"	SA-105	12.000	10.000	...	...	...
K3-2"	SA-105	12.000	10.000	...	...	...
K4-2"	SA-105	12.000	10.000	...	...	...
K1A-2"	SA-105	12.000	10.000	...	...	...
K1B-2"	SA-105	12.000	10.000	...	...	...
N3-4"	SA-106 B	12.000	10.000	...	...	...
N4-4"	SA-106 B	12.000	10.000	...	...	...
N5-4"	SA-106 B	12.000	10.000	...	...	...
N6-4"	SA-106 B	12.000	10.000	...	...	...
N2-8"	SA-106 B	12.000	10.000	...	...	...
N1-18	SA-516 70	12.000	10.000	...	...	...
M1-24"	SA-516 70	10.000	10.000	12.000	12.000	...
M2-24"	SA-516 70	10.000	10.000	12.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K2B-2"	650.000	180.0	206.38	0.00	Shell-1
K2A-2"	450.000	0.0	477.65	0.00	Shell-1
K3-2"	3450.000	0.0	430.15	0.00	Shell12
K4-2"	4050.000	0.0	477.65	0.00	Shell12
K1A-2"	9549.999	0.0	477.65	0.00	shell14
K1B-2"	9450.000	180.0	206.38	0.00	shell14
N3-4"	1050.000	0.0	414.15	0.00	Shell-1
N4-4"	2550.000	0.0	468.25	0.00	Shell-1

CCW Expansion Vessel (V8804)

PV Elite 28

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Nozzle Schedule:

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HEXAGON

N5-4"	7249.999	0.0	414.15	0.00	Shell3
N6-4"	7649.999	0.0	414.15	0.00	Shell3
N2-8"	6549.999	180.0	200.00	0.00	N1-18
N1-18	7049.999	180.0	750.00	0.00	Shell3
M1-24"	...	0.0	250.00	0.00	left Dish
M2-24"	...	0.0	250.00	0.00	Right dish

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**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 4000.0mm. ID X 50.0mm.	SA-516 70
2	SADDLE: 250.0mm X 150 DEG	SA-516 70
1	PLATFORM: 1000mm X 4024mm WIDE	
1	LADDER: 6600.0mm	...
2	BAR RING STIFFENER: 4024.0mm ID X 4374.0mm OD X 12.0mm	SA-516 70
4	LIFTING LUGS:	SA-516 70
4	LIFTING LUG PADS: 350.0mm X 450.0mm X 12.0mm THK	
1	BAR RING STIFFENER: 4024.0mm ID X 4324.0mm OD X 12.0mm	SA-516 70
1	BAR RING STIFFENER: 4020.0mm ID X 4320.0mm OD X 12.0mm	SA-516 70
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
9	CYLINDRICAL SEGMENT 12.0mm. THK X 2214.3mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 1494.8mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 10351.9mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 6642.8mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm. THK X 2214.3mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 10351.9mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 6642.8mm. X 1900.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm. THK X 2214.3mm. X 2500.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 10351.9mm. X 2500.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 6642.8mm. X 2500.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
left Dish	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
left Dish	[7]	B	-23	-40	-29	0.689	12.000	0.85	No
Shell-1	[8]	B	-23	-40	-29	0.689	12.000	0.85	No
Shell2	[8]	B	-23	-40	-29	0.689	12.000	0.85	No
Shell3	[8]	B	-23	-40	-29	0.689	12.000	0.85	No
shell4	[8]	B	-23	-40	-29	0.689	12.000	0.85	No
Right dish	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
Right dish	[7]	B	-23	-40	-29	0.689	12.000	0.85	No
Ring R1 Fr20	[23]	B	-23	-40	-29	0.689	12.000	0.85	No
Ring R1 Fr30	[23]	B	-23	-40	-29	0.689	12.000	0.85	No
Ring R1 Fr40	[23]	B	-23	-40	-29	0.689	12.000	0.85	No
Ring R1 Fr50	[23]	B	-23	-40	-29	0.689	12.000	0.85	No
Left Saddle	[24]	B	-23	-40	-29	0.689	12.000	0.85	No
New Saddle	[24]	B	-23	-40	-29	0.689	12.000	0.85	No
M1-24"	[1]	B	-29	-44	-29	0.735	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B-2"	[1]	A	-3	-29	-29	0.688	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A-2"	[1]	A	-3	-29	-29	0.638	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-4"	[1]	B	-3	-104	-29	0.022	11.801	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-4"	[1]	B	-3	-104	-29	0.022	11.801	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2"	[1]	A	-3	-29	-29	0.638	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4-2"	[1]	A	-3	-29	-29	0.638	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-18	[1]	B	-7	-48	-29	0.389	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2-8"	[1]	B	6	-104		0.032	15.980	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-4"	[1]	B	-3	-104	-29	0.022	11.801	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-4"	[1]	B	-3	-104	-29	0.022	11.801	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2"	[1]	A	-3	-29	-29	0.638	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B-2"	[1]	A	-3	-29	-29	0.688	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M2-24"	[1]	B	-29	-44	-29	0.735	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: 6 -29

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable



- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

CCW Expansion Vessel (V8804)

Design Internal Pressure (for Hydrotest)	3.5	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	Air/Water/Steam	
Degree of Radiography	RT-3	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	21500 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.790
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	left Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	4000 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.5171 bars
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3700 mm.
Liquid Density	0.0009874 kg./cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	10
Detail Type	Weight
Detail ID	Davit-1
Dist. from "FROM" Node / Offset dist	-900 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	300 mm.

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm.



Inside Diameter	4000	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.5171	bars
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2100	mm.
Width of Saddle	250	mm.
Height of Saddle at Bottom	2260	mm.
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	500	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	162.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	105000	cm ²
Platform Weight	7430.7	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	4024	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	150	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	-2000	mm.
Ladder End Elevation	4600	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	10500	mm.

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3700	mm.
Liquid Density	0.0009874	kg./cm ³

Element From Node	20	
Detail Type	Ring	
Detail ID	Ring R1 Fr20	
Dist. from "FROM" Node / Offset dist	1263	mm.
Inside Diameter of Ring	4024	mm.
Thickness of Ring	12	mm.
Outside Diameter of Ring	4374	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.



Using Custom Stiffener Section	No
Element From Node	20
Detail Type	Nozzle
Detail ID	K2B-2"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K2A-2"
Dist. from "FROM" Node / Offset dist	400 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	N3-4"
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Nozzle
Detail ID	N4-4"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell12
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	4000 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	82 °C



Design External Pressure	0.5171	bars
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3700	mm.
Liquid Density	0.0009874	kg./cm³

Element From Node	30	
Detail Type	Ring	
Detail ID	Ring R1 Fr30	
Dist. from "FROM" Node / Offset dist	888	mm.
Inside Diameter of Ring	4024	mm.
Thickness of Ring	12	mm.
Outside Diameter of Ring	4374	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-2"	
Dist. from "FROM" Node / Offset dist	400	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	30	
Detail Type	Weight	
Detail ID	Clip	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	110	Kgf
Offset from Element Centerline	0	mm.

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell3	
Distance "FROM" to "TO"	1900	mm.
Inside Diameter	4000	mm.
Element Thickness	12	mm.



Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.5171	bars
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3700	mm.
Liquid Density	0.0009874	kg./cm ³
Element From Node	40	
Detail Type	Ring	
Detail ID	Ring R1 Fr40	
Dist. from "FROM" Node / Offset dist	510	mm.
Inside Diameter of Ring	4024	mm.
Thickness of Ring	12	mm.
Outside Diameter of Ring	4324	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1-18	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-8"	
Dist. from "FROM" Node / Offset dist	500	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N5-4"	
Dist. from "FROM" Node / Offset dist	1200	mm.
Nozzle Diameter	4	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	40
Detail Type	Nozzle
Detail ID	N6-4"
Dist. from "FROM" Node / Offset dist	1600 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	5/6
Element From Node	50
Element To Node	70
Element Type	Cylinder
Description	shell4
Distance "FROM" to "TO"	2500 mm.
Inside Diameter	4000 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.5171 bars
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	400 mm.
Width of Saddle	250 mm.
Height of Saddle at Bottom	2260 mm.
Saddle Contact Angle	150.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	500 mm.
Thickness of Wear Plate	12 mm.
Contact Angle, Wear Plate (degrees)	162.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3700 mm.
Liquid Density	0.0009874 kg./cm ³

Element From Node	50
Detail Type	Ring
Detail ID	Ring R1 Fr50
Dist. from "FROM" Node / Offset dist	1335 mm.
Inside Diameter of Ring	4020 mm.
Thickness of Ring	12 mm.
Outside Diameter of Ring	4320 mm.



Material Name SA-516 70
Height of Section Ring 0 mm.
Using Custom Stiffener Section No

Element From Node 50
Detail Type Nozzle
Detail ID K1A-2"
Dist. from "FROM" Node / Offset dist 1600 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-105

Element From Node 50
Detail Type Nozzle
Detail ID K1B-2"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 180.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-105

Element# 6/6
Element From Node 70
Element To Node 80
Element Type Elliptical
Description Right dish
Distance "FROM" to "TO" 50 mm.
Inside Diameter 4000 mm.
Element Thickness 10 mm.
Internal Corrosion Allowance 1.6 mm.
Nominal Thickness 12 mm.
External Corrosion Allowance 0 mm.
Design Internal Pressure 3.5 bars
Design Temperature Internal Pressure 82 °C
Design External Pressure 0.5171 bars
Design Temperature External Pressure 82 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 0.85
Efficiency, Circumferential Seam 0.85
Elliptical Head Factor 2.0
Weld is pre-Heated No

Element From Node 70
Detail Type Liquid
Detail ID Liquid 70
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Liquid 3700 mm.
Liquid Density 0.0009874 kg./cm³

Element From Node 70
Detail Type Nozzle
Detail ID M2-24"
Dist. from "FROM" Node / Offset dist 0 mm.
Nozzle Diameter 24 in.
Nozzle Schedule None
Nozzle Class 150
Layout Angle 0.0



Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	70
Detail Type	Weight
Detail ID	Davit-2
Dist. from "FROM" Node / Offset dist	900 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	300 mm.

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
left Dish		50	...	50	...
Shell-1		3050	...	3000	...
Shell2		6050	...	3000	...
Shell3		7950	...	1900	...
shell4		10450	...	2500	...
Right dish		10500	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
left Dish		3.8588	12	1.6	4000	117.3	1
Shell-1		3.8588	12	1.6	4000	117.3	2
Shell2		3.8588	12	1.6	4000	117.3	3
Shell3		3.8588	12	1.6	4000	117.3	4
shell4		3.8588	12	1.6	4000	117.3	5
Right dish		3.8588	12	1.6	4000	117.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
left Dish		3.5	4.56663	5.86162	10	8.18035	1
Shell-1		3.5	5.71649	7.01222	12	8.19822	2
Shell2		3.5	5.71649	7.01222	12	8.19822	3
Shell3		3.5	5.71649	7.01222	12	8.19822	4
shell4		3.5	5.71649	7.01222	12	8.19822	5
Right dish		3.5	4.56663	5.86162	10	8.18035	6

Minimum 5 6

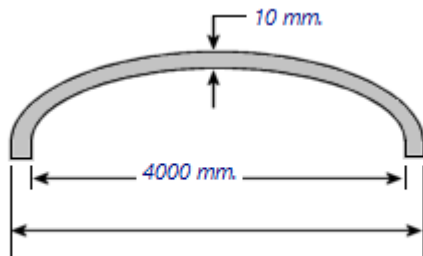
MAWP: 5 bars, limited by: Right dish.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: left Dish

Elliptical Head Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope) S		138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb) Sa		138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		108.1 N./mm ²
Spec. Min. Thk t,e, or sa		10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		181846.0 cm ²	SG of Contents		0.989
Element Volume		9007512.0 cm ³	Weight of Contents		8760.1 Kg
Empty Weight		1850.3 Kg	Operating Weight		10610.5 Kg
			Comp. All. Ope. Stress		62.7 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 50.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 25.000 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.859 \cdot 4003.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 3.859)$$

$$= 6.5803 + 1.6000 = 8.1803 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 8.4) / (0.999 \cdot 4003.2002 + 0.2 \cdot 8.4)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 4000.0 + 0.2 \cdot 10.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.859 \cdot (0.999 \cdot 4003.2002 + 0.2 \cdot 8.4)) / (2 \cdot 0.85 \cdot 8.4)$$

$$= 108.114 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 3.859) + 1.6$$

$$= 8.198 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4)$$

$$= 6 - 0 = 6 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (4003.2 / (2 \cdot 1001.6)))^2 / 6$$

$$= 0.998936$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.312 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 8.4$, $c = 1.6$ mm., $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 50.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 25.000 mm.

EXTERNAL PRESSURE RESULTS for: left Dish

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.400	4020.00	478.57	0.0002902	29.02

$$MAEP = B / (K_0 \cdot D_o / t) = 29.015 / (0.9 \cdot 478.5714) = 0.6736 \text{ bars}$$

CCW Expansion Vessel (V8804)

PV Elite 28

Licensee: SL User



FileName : CCW Expansion Vessel (V8804)-7%

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Element: left Dish

Step: 4 3:03pm Sep 16,2026



HEXAGON

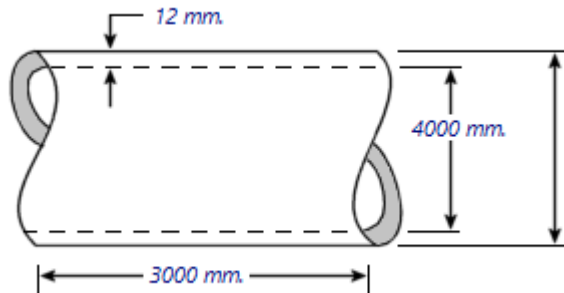
Results for Required Thickness

Tca

Tca	Outer Dia	Do/t	Factor A	Factor B
7.360	4020.00	546.19	0.0002543	25.42

$$MAEP = B / (K0 * Do / t) = 25.4227 / (0.9 * 546.1949) = 0.5171 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-1**

Cylindrical Shell Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		87.7 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		379253.1 cm ²	SG of Contents		0.989
Element Volume		37705864.0 cm ³	Weight of Contents		35997.8 Kg
Empty Weight		3516.8 Kg	Operating Weight		39514.6 Kg
			Comp. All. Ope. Stress		77.5 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 3.859) \\
 &= 6.5982 + 1.6000 = 8.1982 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4) \\
 &= 6 - 0 = 6 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2000.0 + 0.6 \cdot 12.0) \\
 &= 7 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.859 \cdot (2001.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 87.651 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.299 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.689	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to Ring R1 Fr20 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	1646.33	386.92	0.4091	0.0004450	44.49

$$MAEP = (4*B)/(3*(Do/t)) = (4*44.4871)/(3*386.9231) = 1.5329 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.785	4024.00	1646.33	593.10	0.4091	0.0002301	23.00

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.0047)/(3*593.0969) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4667.51	386.92	1.1599	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0184)/(3*386.9231) = 0.5175 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr20 to 30 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2625.00	386.92	0.6523	0.0002723	27.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.2221)/(3*386.9231) = 0.938 \text{ bars}$$

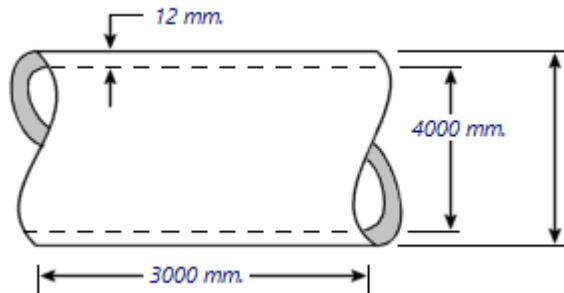
Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.228	4024.00	2625.00	489.08	0.6523	0.0001897	18.97



$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9697)/(3*489.0839) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4666.98	386.92	1.1598	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0201)/(3*386.9231) = 0.5176 \text{ bars}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell2**

Cylindrical Shell Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		87.7 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		379253.1 cm ²	SG of Contents		0.989
Element Volume		37705864.0 cm ³	Weight of Contents		35997.8 Kg
Empty Weight		3516.8 Kg	Operating Weight		39514.6 Kg
			Comp. All. Ope. Stress		77.5 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 3.859) \\
 &= 6.5982 + 1.6000 = 8.1982 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4) \\
 &= 6 - 0 = 6 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2000.0 + 0.6 \cdot 12.0) \\
 &= 7 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.859 \cdot (2001.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 87.651 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.299 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.689	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell2

Cylindrical Shell From 30 to Ring R1 Fr30 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2625.00	386.92	0.6523	0.0002723	27.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.2221)/(3*386.9231) = 0.938 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.228	4024.00	2625.00	489.08	0.6523	0.0001897	18.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9697)/(3*489.0839) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4666.98	386.92	1.1598	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0201)/(3*386.9231) = 0.5176 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2622.00	386.92	0.6516	0.0002726	27.25

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.2547)/(3*386.9231) = 0.9391 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.224	4024.00	2622.00	489.30	0.6516	0.0001898	18.98

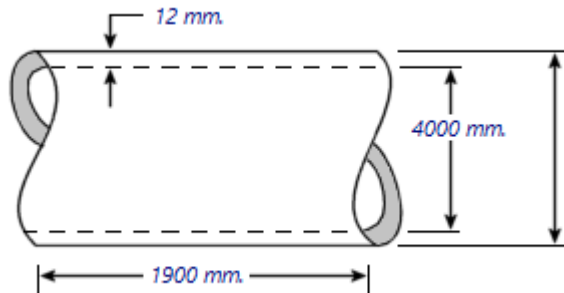


$$MAEP = (4*B)/(3*(Do/t)) = (4*18.978)/(3*489.2968) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4666.98	386.92	1.1598	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0201)/(3*386.9231) = 0.5176 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell3**

Cylindrical Shell Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		87.7 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		240193.6 cm ²	SG of Contents		0.989
Element Volume		23880380.0 cm ³	Weight of Contents		22798.6 Kg
Empty Weight		2227.3 Kg	Operating Weight		25025.9 Kg
			Comp. All. Ope. Stress		77.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 3.859) \\
 &= 6.5982 + 1.6000 = 8.1982 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4) \\
 &= 6 - 0 = 6 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2000.0 + 0.6 \cdot 12.0) \\
 &= 7 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.859 \cdot (2001.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 87.651 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.299 %

Minimum Design Metal Temperature Results:

Govern. thk, tg = 12.0, tr = 8.427, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.689, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.689	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell3

Cylindrical Shell From 40 to Ring R1 Fr40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2622.00	386.92	0.6516	0.0002726	27.25

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.2547)/(3*386.9231) = 0.9391 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.224	4024.00	2622.00	489.30	0.6516	0.0001898	18.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.978)/(3*489.2968) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4666.98	386.92	1.1598	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0201)/(3*386.9231) = 0.5176 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr40 to 50 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2725.00	386.92	0.6772	0.0002618	26.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.1755)/(3*386.9231) = 0.902 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.347	4024.00	2725.00	482.11	0.6772	0.0001870	18.70

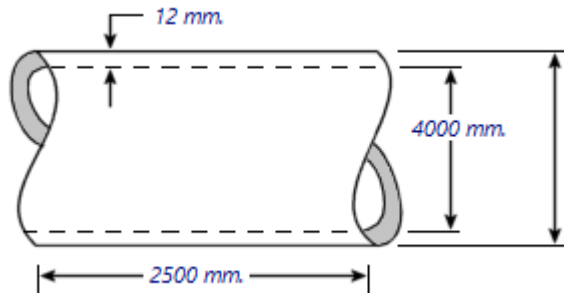


$$MAEP = (4*B)/(3*(Do/t)) = (4*18.6994)/(3*482.1101) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4667.24	386.92	1.1599	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0192)/(3*386.9231) = 0.5175 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: shell4**

Cylindrical Shell Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		2500.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		87.7 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		316044.2 cm ²	SG of Contents		0.989
Element Volume		31421556.0 cm ³	Weight of Contents		29998.2 Kg
Empty Weight		2930.6 Kg	Operating Weight		32928.8 Kg
			Comp. All. Ope. Stress		77.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 3.859) \\
 &= 6.5982 + 1.6000 = 8.1982 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4) \\
 &= 6 - 0 = 6 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 12.0) / (2000.0 + 0.6 \cdot 12.0) \\
 &= 7 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.859 \cdot (2001.6001 + 0.6 \cdot 10.4)) / (0.85 \cdot 10.4) \\
 &= 87.651 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.299 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.689	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: shell4

Cylindrical Shell From 50 to Ring R1 Fr50 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	2725.00	386.92	0.6772	0.0002618	26.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.1755)/(3*386.9231) = 0.902 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.347	4024.00	2725.00	482.11	0.6772	0.0001870	18.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.6994)/(3*482.1101) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4667.24	386.92	1.1599	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.0192)/(3*386.9231) = 0.5175 \text{ bars}$$

Cylindrical Shell From Ring R1 Fr50 to 70 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	1548.33	386.92	0.3848	0.0004754	47.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*47.5314)/(3*386.9231) = 1.6378 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.608	4024.00	1548.33	608.92	0.3848	0.0002362	23.62



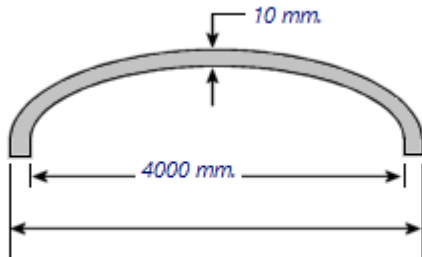
$$MAEP = (4*B)/(3*(Do/t)) = (4*23.6181)/(3*608.9218) = 0.5171 \text{ bars}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.400	4024.00	4667.32	386.92	1.1599	0.0001502	15.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.019)/(3*386.9231) = 0.5175 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right dish

Elliptical Head Design Information:					
Design Pressure	P	3.859 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	4000.000 mm.	Actual Stress		108.1 N./mm ²
Spec. Min. Thk	t,e, or sa	10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		181846.0 cm ²	SG of Contents		0.989
Element Volume		9007512.0 cm ³	Weight of Contents		8760.1 Kg _f
Empty Weight		1850.3 Kg _f	Operating Weight		10610.5 Kg _f
			Comp. All. Ope. Stress		77.5 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 50.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 25.000 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.859 \cdot 4003.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 3.859)$$

$$= 6.5803 + 1.6000 = 8.1803 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 8.4) / (0.999 \cdot 4003.2002 + 0.2 \cdot 8.4)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 4000.0 + 0.2 \cdot 10.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.859 \cdot (0.999 \cdot 4003.2002 + 0.2 \cdot 8.4)) / (2 \cdot 0.85 \cdot 8.4)$$

$$= 108.114 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.859 \cdot 2001.6001) / (138.0 \cdot 0.85 - 0.6 \cdot 3.859) + 1.6$$

$$= 8.198 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.359 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.4) / (2001.6001 + 0.6 \cdot 10.4)$$

$$= 6 - 0 = 6 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (4003.2 / (2 \cdot 1001.6)))^2 / 6$$

$$= 0.998936$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.312 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 8.4$, $c = 1.6 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4.925 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 50.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 25.000 mm.

EXTERNAL PRESSURE RESULTS for: Right dish

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.400	4020.00	478.57	0.0002902	29.02

$$MAEP = B / (K_0 \cdot D_o / t) = 29.015 / (0.9 \cdot 478.5714) = 0.6736 \text{ bars}$$

CCW Expansion Vessel (V8804)

PV Elite 28

Licensee: SL User



FileName : CCW Expansion Vessel (V8804)-7%

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Element: Right dish

Step: 9 3:03pm Sep 16,2026



HEXAGON

Results for Required Thickness

Tca

Tca	Outer Dia	Do/t	Factor A	Factor B
7.360	4020.00	546.19	0.0002543	25.42

$$MAEP = B / (K0 * Do / t) = 25.4227 / (0.9 * 546.1949) = 0.5171 \text{ bars}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.937 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7.130 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	5.023 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	4.567 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 3.5 • 1.0

= 4.550 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
left Dish	117.5	234.0	0.502	4.99
Shell-1	98.2	234.0	0.420	4.99
Shell12	98.2	234.0	0.420	4.99
Shell13	98.2	234.0	0.420	4.99
shell14	98.2	234.0	0.420	4.99
Right dish	117.5	234.0	0.502	4.99

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
N3-4"	Nozzle	118.00	118.00	1.000
N4-4"	Nozzle	118.00	118.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N1-18	Nozzle	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N6-4"	Nozzle	118.00	118.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	138.00	1.000
Ring R1 Fr30	138.00	138.00	1.000
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr50	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



Description	Ambient	Operating	Ratio
left Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
Right dish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1-24"	5.23	179.40	0.029
K2B-2"	1.05	179.40	0.006
K2A-2"	1.05	179.40	0.006
N3-4"	2.22	153.40	0.014
N4-4"	2.22	153.40	0.014
K3-2"	1.05	179.40	0.006
K4-2"	1.05	179.40	0.006
N1-18	5.51	179.40	0.031
N2-8"	3.22	153.40	0.021
N5-4"	2.22	153.40	0.014
N6-4"	2.22	153.40	0.014
K1A-2"	1.05	179.40	0.006
K1B-2"	1.05	179.40	0.006
M2-24"	5.23	179.40	0.029

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.937 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7.130 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	5.023 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	4.567 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 3.5 • 1.0

= 4.550 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
left Dish	139.8	234.0	0.598	4.99
Shell-1	113.4	234.0	0.484	4.99
Shell12	113.4	234.0	0.484	4.99
Shell13	113.4	234.0	0.484	4.99
shell14	113.4	234.0	0.484	4.99
Right dish	139.8	234.0	0.598	4.99

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
N3-4"	Nozzle	118.00	118.00	1.000
N4-4"	Nozzle	118.00	118.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N1-18	Nozzle	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N6-4"	Nozzle	118.00	118.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	138.00	1.000
Ring R1 Fr30	138.00	138.00	1.000
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr50	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



Description	Ambient	Operating	Ratio
left Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
Right dish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1-24"	5.23	179.40	0.029
K2B-2"	1.05	179.40	0.006
K2A-2"	1.05	179.40	0.006
N3-4"	2.22	153.40	0.014
N4-4"	2.22	153.40	0.014
K3-2"	1.05	179.40	0.006
K4-2"	1.05	179.40	0.006
N1-18	5.51	179.40	0.031
N2-8"	3.22	153.40	0.021
N5-4"	2.22	153.40	0.014
N6-4"	2.22	153.40	0.014
K1A-2"	1.05	179.40	0.006
K1B-2"	1.05	179.40	0.006
M2-24"	5.23	179.40	0.029

Elements Suitable for Test Pressure.

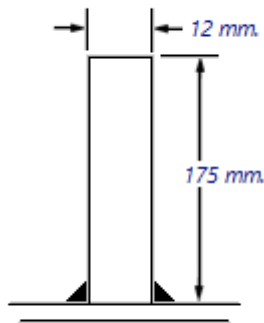
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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****MDMT Calculations for: Ring R1 Fr20, Shell to Ring Junction, Curve: B**Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$ Thickness Ratio = $t_r * E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr20, SA-516 70 , Bar Ring: 175.0 x 12.0 mm.**Material treatment options:**

Effective Length of Shell: 225 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	23.403	5.2000	121.696
Ring :	21.000	97.9000	2055.899
Total:	44.403		2177.595

Centroid of Ring plus Shell: 49 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.109	43.8416	44982.629
Ring :	535.937	-48.8584	50129.945
Total:	538.046		95112.578

Available Moment of Inertia, Ring + Shell: 0.149E+04 cm⁴Required Stress in Ring plus Shell Breq 13.71 N./mm²

Required Strain in Ring plus Shell Areq 0.0001370

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

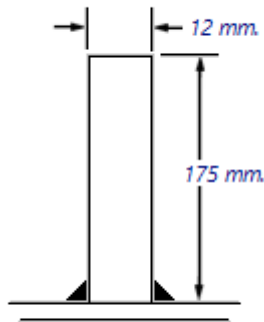
$$= (4024.0^2 \cdot 2135.6667(10.4 + 21.0/2135.6667)0.000137) / 10.9$$

$$= 495 \text{ cm}^4$$

MDMT Calculations for: Ring R1 Fr30, Shell to Ring Junction, Curve: BGovrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$ Thickness Ratio = $t_r * E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr30, SA-516 70 , Bar Ring: 175.0 x 12.0 mm.
Material treatment options:


Effective Length of Shell: 225 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	23.403	5.2000	121.696
Ring :	21.000	97.9000	2055.899
Total:	44.403		2177.595

Centroid of Ring plus Shell: 49 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.109	43.8416	44982.629
Ring :	535.937	-48.8584	50129.945
Total:	538.046		95112.578

Available Moment of Inertia, Ring + Shell: 0.149E+04 cm⁴

Required Stress in Ring plus Shell Breq 13.93 N./mm²
 Required Strain in Ring plus Shell Areq 0.0001390

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4024.0^2 \cdot 2623.4998(10.4 + 21.0/2623.4998)0.000139) / 10.9$$

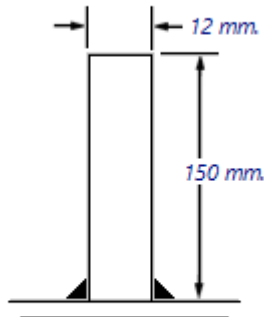
$$= 607 \text{ cm}^4$$

MDMT Calculations for: Ring R1 Fr40, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr40, SA-516 70 , Bar Ring: 150.0 x 12.0 mm.
Material treatment options:



Effective Length of Shell: 225 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	23.403	5.2000	121.696
Ring :	18.000	85.4000	1537.199
Total:	41.403		1658.895

Centroid of Ring plus Shell: 40 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.109	34.8670	28451.242
Ring :	337.499	-45.3330	36991.383
Total:	339.609		65442.625

Available Moment of Inertia, Ring + Shell: 0.994E+03 cm⁴

Required Stress in Ring plus Shell Breq 14.09 N./mm²
 Required Strain in Ring plus Shell Areq 0.0001410

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4024.0^2 \cdot 2673.5002(10.4 + 18.0/2673.5002)0.000141) / 10.9$$

$$= 620 \text{ cm}^4$$

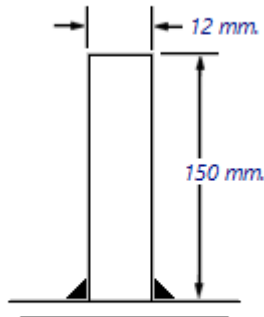
MDMT Calculations for: Ring R1 Fr50, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.689$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr50, SA-516 70 , Bar Ring: 150.0 x 12.0 mm.

Material treatment options:



Effective Length of Shell: 225 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	23.403	5.2000	121.696
Ring :	18.000	85.4000	1537.199
Total:	41.403		1658.895

Centroid of Ring plus Shell: 40 mm.

	Inertia	Distance	A*Dist ²
Shell:	2.109	34.8670	28451.242
Ring :	337.499	-45.3330	36991.383
Total:	339.609		65442.625

Available Moment of Inertia, Ring + Shell: 0.994E+03 cm⁴

Required Stress in Ring plus Shell Breq 13.88 N./mm²

Required Strain in Ring plus Shell Areq 0.0001390

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4024.0^2 \cdot 2136.6665(10.4 + 18.0/2136.6665)0.000139) / 10.9$$

$$= 496 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	4020	8.4	0.00029022	29.015
20	Ring	1646.33	4024	10.4	0.00044497	44.4871
Ring	30	2625	4024	10.4	0.00027228	27.2221
30	Ring	2625	4024	10.4	0.00027228	27.2221
Ring	40	2622	4024	10.4	0.00027261	27.2547
40	Ring	2622	4024	10.4	0.00027261	27.2547
Ring	50	2725	4024	10.4	0.00026181	26.1755
50	Ring	2725	4024	10.4	0.00026181	26.1755
Ring	70	1548.33	4024	10.4	0.00047542	47.5314
70	80	No Calc	4020	8.4	0.00029022	29.015

External Pressure Calculations:

From	To	External Actual T.	External Required T.	External Design Pressure	External M.A.W.P.
------	----	--------------------	----------------------	--------------------------	-------------------



		mm.	mm.	bars	bars
10	20	10	8.96001	0.51711	0.67361
20	Ring	12	8.38473	0.51711	1.53293
Ring	30	12	9.82763	0.51711	0.93802
30	Ring	12	9.82763	0.51711	0.93802
Ring	40	12	9.82405	0.51711	0.93914
40	Ring	12	9.82405	0.51711	0.93914
Ring	50	12	9.94664	0.51711	0.90195
50	Ring	12	9.94664	0.51711	0.90195
Ring	70	12	8.2084	0.51711	1.63783
70	80	10	8.96001	0.51711	0.67361

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	1646.33	4667.51	No Calc	No Calc
Ring	30	2625	4666.98	494.779	1489.17
30	Ring	2625	4666.98	No Calc	No Calc
Ring	40	2622	4666.98	606.765	1489.17
40	Ring	2622	4666.98	No Calc	No Calc
Ring	50	2725	4667.24	620.103	994.034
50	Ring	2725	4667.24	No Calc	No Calc
Ring	70	1548.33	4667.32	496.022	994.034
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	1850.35	9007512	1603.63	9028646	129.524
20	30	3516.79	37705864	3049.05	37766224	246.175
30	40	3516.79	37705864	3049.05	37766224	246.175
40	50	2227.29	23880380	1931.06	23918608	155.91
50	70	2930.65	31421556	2540.87	31471856	205.145
70	80	1850.35	9007512	1603.63	9028646	129.524
Total		15892	0.149E+09	13777	0.149E+09	1112

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	8760.11	-333.333	150	...	Liquid 10
10	Nozl	486.018	-1000	...	...	M1-24"
10	Wght	50	-1000	...	300	Davit-1
20	Sadl	1601.44	2100	2118	...	Left Saddle
20	Plat	7950.88	5250	2088.2	...	PLAT:[1 OF 1]
20	Liqd	35997.8	1500	150	...	Liquid 20
20	Ring	229.724	1263	...	...	Ring R1 Fr20
20	Nozl	11.0391	600	-2025.4	...	K2B-2"
20	Nozl	19.0652	400	2025.4	...	K2A-2"
20	Nozl	27.1728	1000	2057.15	...	N3-4"
20	Nozl	28.8778	2500	2057.15	...	N4-4"
30	Liqd	35997.8	1500	150	...	Liquid 30
30	Ring	229.724	888	...	...	Ring R1 Fr30
30	Nozl	17.6598	400	2025.4	...	K3-2"
30	Nozl	19.0652	1000	2025.4	...	K4-2"
30	Wght	110	...	...	...	Clip
40	Liqd	22798.6	950	150	...	Liquid 40
40	Ring	195.734	510	...	...	Ring R1 Fr40
40	Nozl	254.999	1000	-2228.6	...	N1-18
40	Nozl	38.7038	500	-2109.54	...	N2-8"
40	Nozl	27.1728	1200	2057.15	...	N5-4"
40	Nozl	27.1728	1600	2057.15	...	N6-4"
50	Sadl	1601.44	400	2118	...	New Saddle
50	Liqd	29998.2	1250	150	...	Liquid 50
50	Ring	195.546	1335	...	...	Ring R1 Fr50
50	Nozl	19.0652	1600	2025.4	...	K1A-2"
50	Nozl	11.0391	1500	-2025.4	...	K1B-2"
70	Liqd	8760.11	383.333	150	...	Liquid 70
70	Nozl	486.018	1050	...	...	M2-24"
70	Wght	50	1050	...	300	Davit-2

Total Weight of Each Detail Type:

Saddles	3202.9
Platforms	7950.9
Liquid	142312.7
Stiffeners	850.7
Nozzles	1473.1
Weights	210.0

Sum of the Detail Weights 156000.3 kg.

Weight Summation Results: (kg.)

Fabricated	Shop Test	Shipping	Erected	Empty	Operating
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Main Elements	17004.7	17004.7	17004.7	17004.7	17004.7	17004.7
Saddles	3202.9	3202.9	3202.9	3202.9	3202.9	3202.9
Stif. Rings	850.7	850.7	850.7	850.7	850.7	850.7
Nozzles	1473.1	1473.1	1473.1	1473.1	1473.1	1473.1
Wld Weights	210.0	210.0	210.0	210.0	210.0	210.0
Platforms	...	...	...	7950.9	7950.9	7950.9
Ope. Liquid	...	...	...	...	...	142312.7
Test Liquid	...	148637.9	...	...	...	...
Totals	22741.4	171379.3	22741.4	30692.2	30692.2	173004.9

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	22741.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	171379.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	22741.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	30692.2 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	30692.2 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	173004.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	170742.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	179330.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	181846	179721
20	30	379253	376991
30	40	379253	376991
40	50	240194	238761
50	70	316044	314159
70	80	181846	179721

Total	1678436.1	1666344.2 cm ²
Total	1806.7	1793.6 ft ²

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**Nozzle Flange MAWP Results: (bars & °C)**

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K2B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K2A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N3-4"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N4-4"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K3-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K4-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N1-18	18.20	19.60	82	150	GR 1.1	...	...	...	...
N2-8"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N5-4"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N6-4"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
M2-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	21500 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.192 \cdot 0.914) / (1 + 1.7 \cdot 3.4 \cdot 0.192)))$$

$$= \min(0.85, 0.883)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.532
Structure Height to Diameter ratio	2.915

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 2260.0 + 21500.0 = 23760.0 \text{ mm.}$$

$$= 77.953 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z $[Kz]$:

$$\text{Because } z (77.953 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Zg)^{2/\alpha}$$

$$= 2.01 (77.953 / 900.0)^{2/9.5}$$

$$= 1.201$$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:

$$= 1, \text{ per Table 26.6-1}$$

As there is No Hill Present: $[Kzt]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.201 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.532 \cdot 36.921$$

$$= 267.1 \text{ lbs. } [121.2] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
left Dish	23760.0	0.000	0.000	0.000	1.201	1.000	78.120
Shell-1	23760.0	0.000	0.000	0.000	1.201	1.000	78.120
Shell2	23760.0	0.000	0.000	0.000	1.201	1.000	78.120
Shell3	23760.0	0.000	0.000	0.000	1.201	1.000	78.120
shell4	23760.0	0.000	0.000	0.000	1.201	1.000	78.120
Right dish	23760.0	0.000	0.000	0.000	1.201	1.000	78.120

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	105000.00	23760.00	78.12	820.26	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Davit-1	0.00	23760.00	78.12	0.00
Clip	0.00	23760.00	78.12	0.00
Davit-2	0.00	23760.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	23760	4824	34300.7	78.12	133.269
20	30	23760	4828.8	144864	78.12	1465.13
30	40	23760	4828.8	144864	78.12	562.84
40	50	23760	4828.8	91747.2	78.12	356.465
50	70	23760	4828.8	120720	78.12	469.034
70	80	23760	4824	34300.7	78.12	133.269

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.790
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(13.97640.75)$
 $= 0.145$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.145, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1.0 \cdot 0.157 \cdot 169802(1 + 2 \cdot 0.79) / (3.0/1.25) \\
 &= 11448.733 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 169802 = 53249.93 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 169802$
 $= 9984.362 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 11448.733 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 169802$
 $= 5325.0 \text{ Kgf}$

**Earthquake Load Calculation:**

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	2000	21225.3	1431.09
20	30	2000	21225.3	1431.09
20	30	2000	21225.3	1431.09
30	40	2000	21225.3	1431.09
40	50	2000	21225.3	1431.09
50	0	2000	21225.3	1431.09
50	70	2000	21225.3	1431.09
70	80	2000	21225.3	1431.09

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5250.000 mm.
Center of Gravity of the Platforms	5300.000 mm.
Center of Gravity of the Liquid	5250.000 mm.
Center of Gravity of the Stiffening Rings	5061.478 mm.
Center of Gravity of the Nozzles	5499.929 mm.
Center of Gravity of the Added Weights (Operating)	4097.618 mm.
Center of Gravity of the Added Weights (Empty)	4097.618 mm.
Center of Gravity of Bare Shell New and Cold	5249.996 mm.
Center of Gravity of Bare Shell Corroded	5249.999 mm.
Vessel CG in the Operating Condition	5252.127 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5248.492 mm.
Vessel CG in the Test Condition	5249.800 mm.

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	10500.00 mm.
Total vessel weight (No Liquid) Twt	30692.23 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	46038.35 Kgf
Distance to the Lug on the Left End	2150.00 mm.
Distance to the Lug on the Right End	8350.00 mm.
Design Reaction force at the Left End Lug	23030.37 Kgf
Design Reaction force at the Right End Lug	23007.98 Kgf
CG distance during field lifting operation	5248.49 mm.
CG distance from the left end lug/trunnion	3098.49 mm.
CG distance from the nearer lifting lug/trunnion	3098.49 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	1.60	2450.00	74.54 (UG-23)
Shear	1.14	1850.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	9975.0	0.8
50.00	2731.7	0.2	10742.3	0.7
3050.00	20652.7	1.3	2595.8	0.2
6050.00	12289.5	0.8	3158.9	0.2
7950.00	17927.1	1.2	8913.7	0.6
10450.00	2424.7	0.2	767.3	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0215 at elevation 2450.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

CCW Expansion Vessel (V8804)

PV Elite 28 Licensee: SL User

FileName : CCW Expansion Vessel (V8804)-7%

Center of Gravity Calculation: Step: 17 3:03pm Sep 16,2026



HEXAGON

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--- Plot data successfully generated ...----

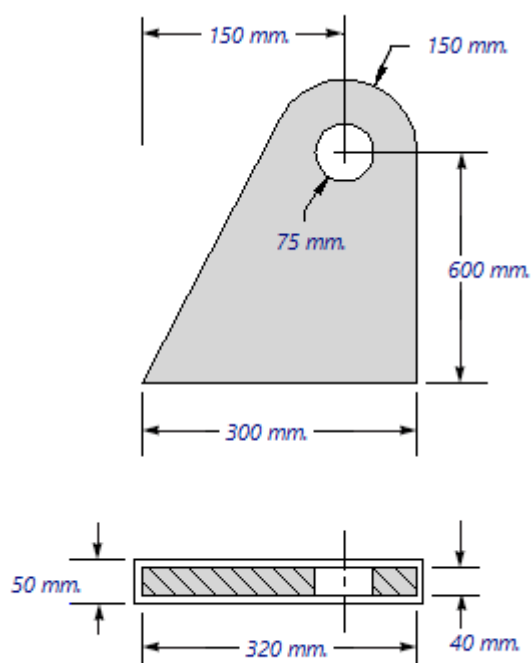
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Lifting Lug Calculations: Lug(s) on Left End of Vessel****Input Values:**

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Total Height of Lifting Lug	w	300.0000 mm.
Thickness of Lifting Lug	t	40.0000 mm.
Diameter of Hole in Lifting Lug	dh	75.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	150.0000 mm.
Height of Lug from Edge to Center of Hole	h	150.0000 mm.
Offset from Vessel Surface to Center of Hole	off	600.0000 mm.
Lug Fillet Weld Size	tw	10.0000 mm.
Length of weld along side of Lifting Lug	wl	320.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	50.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Sling Angle from Horizontal		90.0 degrees
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	350.00 mm.
Reinforcing Pad Longitudinal Length	wpl	450.00 mm.
Reinforcing Pad Thickness	thkp	12.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	11.00 mm.

Lifting Lug Orientation to Vessel: Perpendicular

Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)	30692.23 Kgf
Design reaction force at the left end lug	23030.37 Kgf
Design reaction force at the right end lug	23007.98 Kgf



Force along vessel axis	Fax	0.00 Kgf
Force normal to vessel	Fn	11515.18 Kgf
Force tangential to vessel	Ft	0.00 Kgf

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	Ill	5752.230 cm ⁴
Weld group centroid distance in the long. direction	Yll	167.070 mm.
Dist. of weld group centroid from lug bottom	Yllb	160.000 mm.
Weld group inertia about the circumferential axis	Ilc	267.241 cm ⁴
Weld group centroid distance in the circ. direction	Ylc	27.070 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2)} / ((2 \cdot (wl + wb)) \cdot tw)$$

$$= \sqrt{(0^2 + 0^2 + 11515^2)} / ((2 \cdot (320.0 + 50.0)) \cdot 7.07)$$

$$= 21.58 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot \text{off} \cdot Yll/III) + (Ft \cdot \text{off} \cdot Ylc/IIc)$$

$$= (0 \cdot 600.0 \cdot 167.07/5752.23) +$$

$$(0 \cdot 600.0 \cdot 27.07/267.241)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 21.585 + 0.0$$

$$= 21.58 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2)} / Sha$$

$$= \sqrt{(11515^2 + 0^2)} / 90.0$$

$$= 12.55 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 40.0 + 2 \cdot 0.0$$

$$= 40.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2)} / (Pbw \cdot dh)$$

$$= \sqrt{(11515^2 + 0^2)} / (40.0 \cdot 75.0)$$

$$= 37.64 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot \text{off} / (w \cdot t^2/6) + Fax \cdot \text{off} / (w^2 \cdot t/6)$$

$$= 0 \cdot 600.0 / (300.0 \cdot 40.0^2/6) +$$

$$0 \cdot 600.0 / (300.0^2 \cdot 40.0/6)$$

$$= 0.00 \text{ N./mm}^2$$



Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 11515 / (300.0 \cdot 40.0)$$

$$= 9.41 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 9.4$$

$$= 9.41 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2))^{1/2} / (\pi \cdot \text{Sta}))}$$

$$= \sqrt{((2(11515^2 + 0^2))^{1/2} / (\pi \cdot 105))}$$

$$= 26.1917 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	4000.000 mm.
Vessel Thickness	Tv	12.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	50.00 mm.
Parameter C22	C22	200.00 mm.
Thickness of Reinforcing Pad	Tpad	12.000 mm.
Pad Parameter C11P	C11p	372.000 mm.
Pad Parameter C22P	C22p	461.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

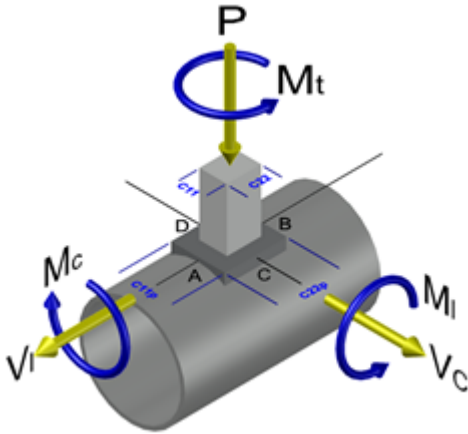
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-11515.2	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-11515.2	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 83.83

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.037	4C	15.337	(A,B)
$N(\Phi) / (P/R_m)$	0.037	3C	16.488	(C,D)
$M(\Phi) / (P)$	0.022	2C1	0.197	(A,B)
$M(\Phi) / (P)$	0.022	1C !	0.206	(C,D)
$N(\Phi) / (MC/(R_m^{**2} * Beta))$	0.020	3A	0.535	(A,B,C,D)
$M(\Phi) / (MC/(R_m^{**2} * Beta))$	0.023	1A !	0.101	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{**2} * Beta))$	0.031	3B	3.975	(A,B,C,D)
$M(\Phi) / (ML/(R_m^{**2} * Beta))$	0.026	1B !	0.062	(A,B,C,D)
$N(x) / (P/R_m)$	0.030	3C	17.294	(A,B)
$N(x) / (P/R_m)$	0.030	4C	15.584	(C,D)
$M(x) / (P)$	0.031	1C1	0.209	(A,B)
$M(x) / (P)$	0.031	2C	0.160	(C,D)
$N(x) / (MC/(R_m^{**2} * Beta))$	0.020	4A	0.554	(A,B,C,D)
$M(x) / (MC/(R_m^{**2} * Beta))$	0.030	2A !	0.061	(A,B,C,D)
$N(x) / (ML/(R_m^{**2} * Beta))$	0.031	4B	1.020	(A,B,C,D)
$M(x) / (ML/(R_m^{**2} * Beta))$	0.036	2B	0.097	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Memb. P	35.9	35.9	35.9	35.9	38.6	38.6	38.6	38.6
Circ. Bend. P	231.2	-231.2	231.2	-231.2	242.7	-242.7	242.7	-242.7
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	267.0	-195.3	267.0	-195.3	281.2	-204.1	281.2	-204.1
Long. Memb. P	40.4	40.4	40.4	40.4	36.4	36.4	36.4	36.4
Long. Bend. P	245.4	-245.4	245.4	-245.4	188.5	-188.5	188.5	-188.5
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	285.8	-204.9	285.8	-204.9	225.0	-152.1	225.0	-152.1
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	285.8	204.9	285.8	204.9	281.2	204.1	281.2	204.1

Dimensionless Parameters used : Gamma = 167.17

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.116	4C	21.762	(A,B)
N(PHI) / (P/Rm)	0.116	3C	12.440	(C,D)
M(PHI) / (P)	0.100	2C1	0.038	(A,B)
M(PHI) / (P)	0.100	1C !	0.071	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.100	3A	6.735	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.103	1A	0.075	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.107	3B	16.436	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.103	1B	0.022	(A,B,C,D)
N(x) / (P/Rm)	0.109	3C	13.431	(A,B)
N(x) / (P/Rm)	0.109	4C	22.446	(C,D)
M(x) / (P)	0.110	1C1	0.064	(A,B)
M(x) / (P)	0.110	2C !	0.038	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.100	4A	12.462	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.115	2A	0.031	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.107	4B	6.840	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.109	2B	0.029	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		102.1	102.1	102.1	102.1	58.4	58.4	58.4	58.4
Circ. Bend. P		180.8	-180.8	180.8	-180.8	333.5	-333.5	333.5	-333.5
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0



Tot. Circ. Str.	282.8	-78.7	282.8	-78.7	391.8	-275.1	391.8	-275.1
Long. Memb. P	63.0	63.0	63.0	63.0	105.3	105.3	105.3	105.3
Long. Bend. P	302.5	-302.5	302.5	-302.5	181.1	-181.1	181.1	-181.1
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	365.5	-239.5	365.5	-239.5	286.4	-75.8	286.4	-75.8
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	365.5	239.5	365.5	239.5	391.8	275.1	391.8	275.1

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		35.9	35.9	35.9	35.9	38.6	38.6	38.6	38.6
Circ. Q (SUS)		231.2	-231.2	231.2	-231.2	242.7	-242.7	242.7	-242.7
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		40.4	40.4	40.4	40.4	36.4	36.4	36.4	36.4
Long. Q (SUS)		245.4	-245.4	245.4	-245.4	188.5	-188.5	188.5	-188.5
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		40.4	40.4	40.4	40.4	38.6	38.6	38.6	38.6
Pm+Pl+Q (Total)		285.8	204.9	285.8	204.9	281.2	204.1	281.2	204.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	40.44	207.00	Passed
Pm+Pl+Q (TOTAL)	285.83	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of	Stress Intensity Values at
---------	----------------------------



Stress Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)	102.1	102.1	102.1	102.1	58.4	58.4	58.4	58.4
Circ. Q (SUS)	180.8	-180.8	180.8	-180.8	333.5	-333.5	333.5	-333.5
Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)	63.0	63.0	63.0	63.0	105.3	105.3	105.3	105.3
Long. Q (SUS)	302.5	-302.5	302.5	-302.5	181.1	-181.1	181.1	-181.1
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	102.1	102.1	102.1	102.1	105.3	105.3	105.3	105.3
Pm+Pl+Q (Total)	365.5	239.5	365.5	239.5	391.8	275.1	391.8	275.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	105.30	207.00	Passed
Pm+Pl+Q (TOTAL)	391.81	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

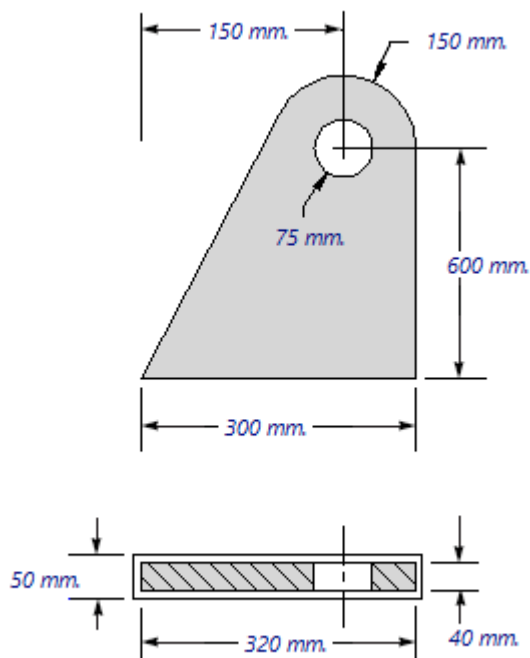
The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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**Lifting Lug Calculations: Lug(s) on Right End of Vessel****Input Values:**

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Total Height of Lifting Lug	w	300.0000 mm.
Thickness of Lifting Lug	t	40.0000 mm.
Diameter of Hole in Lifting Lug	dh	75.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	150.0000 mm.
Height of Lug from Edge to Center of Hole	h	150.0000 mm.
Offset from Vessel Surface to Center of Hole	off	600.0000 mm.
Lug Fillet Weld Size	tw	10.0000 mm.
Length of weld along side of Lifting Lug	wl	320.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	50.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	350.00 mm.
Reinforcing Pad Longitudinal Length	wpl	450.00 mm.
Reinforcing Pad Thickness	thkp	12.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	11.00 mm.

Lifting Lug Orientation to Vessel: Perpendicular
Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)	30692.23 Kgf
Design reaction force at the left end lug	23030.37 Kgf
Design reaction force at the right end lug	23007.98 Kgf



Force along vessel axis	Fax	0.00 Kgf
Force normal to vessel	Fn	11503.99 Kgf
Force tangential to vessel	Ft	0.00 Kgf

Sling angle from horizontal	0.0 deg
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Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I1l	5752.230 cm ⁴
Weld group centroid distance in the long. direction	Y1l	167.070 mm.
Dist. of weld group centroid from lug bottom	Y1lb	160.000 mm.
Weld group inertia about the circumferential axis	I1c	267.241 cm ⁴
Weld group centroid distance in the circ. direction	Y1c	27.070 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2)} / ((2 \cdot (wl + wb)) \cdot tw)$$
$$= \sqrt{(0^2 + 0^2 + 11504^2)} / ((2 \cdot (320.0 + 50.0)) \cdot 7.07)$$
$$= 21.56 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot \text{off} \cdot Y1l/I1l) + (Ft \cdot \text{off} \cdot Y1c/I1c)$$
$$= (0 \cdot 600.0 \cdot 167.07/5752.23) +$$
$$(0 \cdot 600.0 \cdot 27.07/267.241)$$
$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$
$$= 21.564 + 0.0$$
$$= 21.56 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$
$$= 0.4 \cdot 262 \cdot 1.0$$
$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2)} / Sha$$
$$= \sqrt{(11504^2 + 0^2)} / 90.0$$
$$= 12.54 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$
$$= 0.4 \cdot 262 \cdot 1.0$$
$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$
$$= 40.0 + 2 \cdot 0.0$$
$$= 40.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2)} / (Pbw \cdot dh)$$
$$= \sqrt{(11504^2 + 0^2)} / (40.0 \cdot 75.0)$$
$$= 37.61 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$
$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$
$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot \text{off} / (w \cdot t^2/6) + Fax \cdot \text{off} / (w^2 \cdot t/6)$$
$$= 0 \cdot 600.0 / (300.0 \cdot 40.0^2/6) +$$
$$0 \cdot 600.0 / (300.0^2 \cdot 40.0/6)$$
$$= 0.00 \text{ N./mm}^2$$



Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 11504 / (300.0 \cdot 40.0)$$

$$= 9.40 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 9.4$$

$$= 9.40 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2}) / (\pi \cdot \text{Sta}))}$$

$$= \sqrt{((2(11504^2 + 0^2)^{1/2}) / (\pi \cdot 105))}$$

$$= 26.1789 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	4000.000 mm.
Vessel Thickness	Tv	12.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	50.00 mm.
Parameter C22	C22	200.00 mm.
Thickness of Reinforcing Pad	Tpad	12.000 mm.
Pad Parameter C11P	C11p	372.000 mm.
Pad Parameter C22P	C22p	461.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

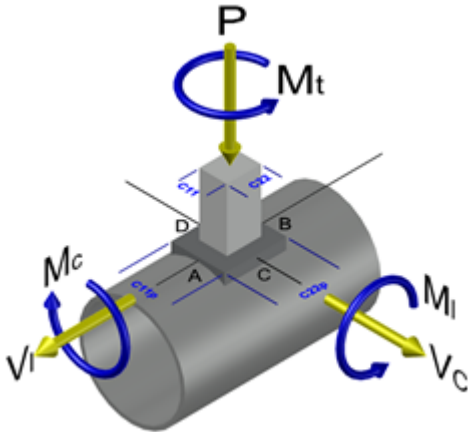
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-11504.0	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-11504.0	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 83.83

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.037	4C	15.337	(A,B)
$N(\Phi) / (P/R_m)$	0.037	3C	16.488	(C,D)
$M(\Phi) / (P)$	0.022	2C1	0.197	(A,B)
$M(\Phi) / (P)$	0.022	1C !	0.206	(C,D)
$N(\Phi) / (MC/(R_m^{**2} * Beta))$	0.020	3A	0.535	(A,B,C,D)
$M(\Phi) / (MC/(R_m^{**2} * Beta))$	0.023	1A !	0.101	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{**2} * Beta))$	0.031	3B	3.975	(A,B,C,D)
$M(\Phi) / (ML/(R_m^{**2} * Beta))$	0.026	1B !	0.062	(A,B,C,D)
$N(x) / (P/R_m)$	0.030	3C	17.294	(A,B)
$N(x) / (P/R_m)$	0.030	4C	15.584	(C,D)
$M(x) / (P)$	0.031	1C1	0.209	(A,B)
$M(x) / (P)$	0.031	2C	0.160	(C,D)
$N(x) / (MC/(R_m^{**2} * Beta))$	0.020	4A	0.554	(A,B,C,D)
$M(x) / (MC/(R_m^{**2} * Beta))$	0.030	2A !	0.061	(A,B,C,D)
$N(x) / (ML/(R_m^{**2} * Beta))$	0.031	4B	1.020	(A,B,C,D)
$M(x) / (ML/(R_m^{**2} * Beta))$	0.036	2B	0.097	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Memb. P	35.8	35.8	35.8	35.8	38.5	38.5	38.5	38.5
Circ. Bend. P	230.9	-230.9	230.9	-230.9	242.4	-242.4	242.4	-242.4
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	266.8	-195.1	266.8	-195.1	281.0	-203.9	281.0	-203.9
Long. Memb. P	40.4	40.4	40.4	40.4	36.4	36.4	36.4	36.4
Long. Bend. P	245.1	-245.1	245.1	-245.1	188.4	-188.4	188.4	-188.4
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	285.6	-204.7	285.6	-204.7	224.8	-151.9	224.8	-151.9
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	285.6	204.7	285.6	204.7	281.0	203.9	281.0	203.9

Dimensionless Parameters used : Gamma = 167.17

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.116	4C	21.762	(A,B)
N(PHI) / (P/Rm)	0.116	3C	12.440	(C,D)
M(PHI) / (P)	0.100	2C1	0.038	(A,B)
M(PHI) / (P)	0.100	1C !	0.071	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.100	3A	6.735	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.103	1A	0.075	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.107	3B	16.436	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.103	1B	0.022	(A,B,C,D)
N(x) / (P/Rm)	0.109	3C	13.431	(A,B)
N(x) / (P/Rm)	0.109	4C	22.446	(C,D)
M(x) / (P)	0.110	1C1	0.064	(A,B)
M(x) / (P)	0.110	2C !	0.038	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.100	4A	12.462	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.115	2A	0.031	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.107	4B	6.840	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.109	2B	0.029	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		102.0	102.0	102.0	102.0	58.3	58.3	58.3	58.3
Circ. Bend. P		180.6	-180.6	180.6	-180.6	333.1	-333.1	333.1	-333.1
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0



Tot. Circ. Str.	282.6	-78.6	282.6	-78.6	391.4	-274.8	391.4	-274.8
Long. Memb. P	62.9	62.9	62.9	62.9	105.2	105.2	105.2	105.2
Long. Bend. P	302.2	-302.2	302.2	-302.2	181.0	-181.0	181.0	-181.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	365.1	-239.2	365.1	-239.2	286.2	-75.8	286.2	-75.8
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	365.1	239.2	365.1	239.2	391.4	274.8	391.4	274.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		35.8	35.8	35.8	35.8	38.5	38.5	38.5	38.5
Circ. Q (SUS)		230.9	-230.9	230.9	-230.9	242.4	-242.4	242.4	-242.4
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		40.4	40.4	40.4	40.4	36.4	36.4	36.4	36.4
Long. Q (SUS)		245.1	-245.1	245.1	-245.1	188.4	-188.4	188.4	-188.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		40.4	40.4	40.4	40.4	38.5	38.5	38.5	38.5
Pm+Pl+Q (Total)		285.6	204.7	285.6	204.7	281.0	203.9	281.0	203.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	40.40	207.00	Passed
Pm+Pl+Q (TOTAL)	285.55	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

	Stress Intensity Values at
--	----------------------------



Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		102.0	102.0	102.0	102.0	58.3	58.3	58.3	58.3
Circ. Q (SUS)		180.6	-180.6	180.6	-180.6	333.1	-333.1	333.1	-333.1
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		62.9	62.9	62.9	62.9	105.2	105.2	105.2	105.2
Long. Q (SUS)		302.2	-302.2	302.2	-302.2	181.0	-181.0	181.0	-181.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		102.0	102.0	102.0	102.0	105.2	105.2	105.2	105.2
Pm+Pl+Q (Total)		365.1	239.2	365.1	239.2	391.4	274.8	391.4	274.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	105.20	207.00	Passed
Pm+Pl+Q (TOTAL)	391.43	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

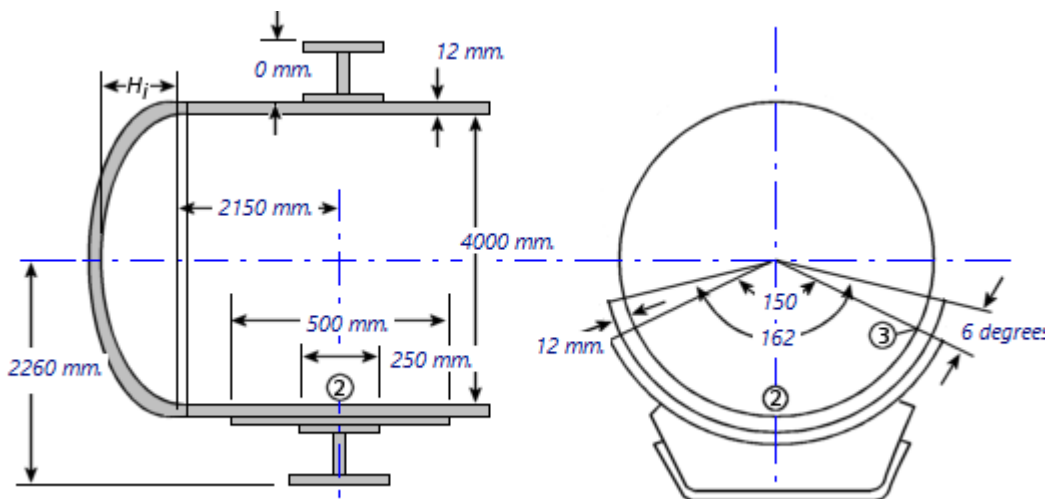
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.80	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.80	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	98468.55	Kgf
Pressure used in Analysis	P 3.665	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	33.12	138.00
Long. Stress at Bottom of Midspan	37.60	138.00
Long. Stress at Top of Saddles	58.50	138.00
Long. Stress at Bottom of Saddles	22.05	138.00
Tangential Shear in Shell	19.35	110.40
Circ. Stress at Horn of Saddle	162.48	172.50
Circ. Compressive Stress in Shell	6.11	138.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (3120.0/2 + 0) \cdot 2260.0/3540.0002$
= 2987.8 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
= $\max(1919.5, 0) \cdot 2260.0/6200.0$
= 699.7 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= $\max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
= $\max(11448.73, 0) \cdot 2260.0/6200.0$
= 4173.2 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (11449/2 + 0) \cdot 2260.0/3540.0002$
= 10963.6 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $87505 + \max(700, 2988, 4173, 10964)$
= 98468.6 Kgf

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 766.08 (4.024 + \pi/4 (4.024 \cdot 1.2)^2)$
= 18823.574 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100070.00	Kgf
Transverse Shear Load	5724.37	Kgf
Longitudinal Shear Load	11448.73	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
= Baseplate Length
= 3540.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
= $-98469 \cdot 2150.0 [1 - (1 - 2150.0/10500.0 + (2006.8^2 - 1005.8^2)/(2 \cdot 2150.0 \cdot 10500.0))/(1 + (4 \cdot 1005.8)/(3 \cdot 10500.0))]$
= -49879.3 Kg-m.

Moment per Equation 4.15.2 [M2]:
= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/(3L)) - 4a/L)$
= $98469 \cdot 10500/4 ((1 + 2(2007^2 - 1006^2)/(10500^2))/(1 + (4 \cdot 1006)/(3 \cdot 10500)) - 4 \cdot 2150/10500)$
= 30037.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
= $P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
= $3.665 \cdot 2006.8/(2 \cdot 10.4) - 30037.6/(\pi \cdot 2006.8^2 \cdot 10.4)$
= 33.12 N./mm²



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 3.665 \cdot 2006.8 / (2 \cdot 10.4) + 30037.6 / (\pi \cdot 2006.8^2 \cdot 10.4)$$

$$= 37.60 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.665 \cdot 2006.8 / (2 \cdot 10.4) - 49879.3 / (0.1607 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= 58.50 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.665 \cdot 2006.8 / (2 \cdot 10.4) + 49879.3 / (0.2792 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= 22.05 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 98469(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.8 / 3))$$

$$= 51558.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 51558.28 / (2006.7999 \cdot 10.4)$$

$$= 19.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2006.8 \cdot 10.4)}$$

$$= 112.684 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(2006.8 \cdot 10.4)}, 2 \cdot 2150.0)$$

$$= 475.37 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.6733 \cdot 98469 \cdot 0.1 / (475.369(10.4 + 1.0 \cdot 12.0))$$

$$= -6.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -98469 / (4(10.4 + 1.0 \cdot 12.0) 475.369) -$$

$$12 \cdot 0.032 \cdot 98469 \cdot 2006.8 / (10500.0(10.4 + 1.0 \cdot 12.0)^2)$$

$$= -162.48 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 6200.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 6200.0(82.0 - 21.1)$$

$$= 4.524 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.
Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C



Saddle Yield Stress	Sy	242.6	N./mm ²
Height of Web at Center	Hw,c	200.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2001.6 \cdot 10.4)} + 500.0)^3) / 12$$

$$= 6.797 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 6.797 + 75.408 (136.719 - 5.2)^2$$

$$= 13050.290 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	725.1	10.4	5.2	75.4	39212.1	0.131E+05
Wearplate	500.0	12.0	16.4	60.0	98400.0	0.869E+04
Web	22.0	200.0	122.4	44.0	538559.6	0.156E+04
BasePlate	476.0	36.0	240.4	171.4	4119493.0	0.186E+05
Totals	...	...	...	350.8	4795664.5	0.419E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4795665 / 350.768$$

$$= 136.719 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 98468.555$$

$$= 25540.0215 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.0960 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2260.0 - 2001.6 \cdot \sin(150.0/2) / 1.309$$

$$= 782.993 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19998.0645 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 63.9821 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(98469 + 1601) \cdot 476.0 / (2 \cdot 3540.0 \cdot 161.733)}$$

$$= 34.984 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$



Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 3910.0679 / (5 - 1)$
 $= 977.517 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 977.517 \cdot 476.0 / 2$
 $= 2326.490 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 100070.008 / (3540.0 \cdot 476.0)$
 $= 5.939 \text{ Kg/cm}^2$

Axial Load [P]:
 $= \text{Ap} \cdot \text{Bp}$
 $= 2326.5 \cdot 5.94$
 $= 13816.401 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 50.16 + 107.527$
 $= 157.687 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / \text{Ar}$
 $= 13816.4 / 157.6869$
 $= 8.593 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 250.0 - 22.0$
 $= 228.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 800456 / 157.687$
 $= 50.762 \text{ mm.}$

Distance to Centroid [C1]:
 $= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$
 $= \max(50.762, 250.0 - 50.762)$
 $= 199.238 \text{ mm.}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$
 $= \sqrt{ (7560.7 / 157.687) }$
 $= 69.244 \text{ mm.}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 2260.0 - \cos(150.0 / 2) (2000.0 + 12.0 + 12.0) - 36.0$
 $= 1700.150 \text{ mm.}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(150.0 / 2) (2000.0 + 12.0 + 12.0)$



= 3910.068 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 3540.0 - 50.8
 = 3489.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (3910.068 - 3489.2)/2
 = 210.434 mm.

Length of Outer Rib [L]:

= $\sqrt{(L_v^2 + D_d^2)}$
 = $\sqrt{(1700.15^2 + 210.434^2)}$
 = 1713.124 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 242.6)}$
 = 127.551

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1713.124/69.244
 = 24.740

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 11448.7 / (2 • 3540.0) • 977.517 • 1713.12/2
 = 1353.991 Kg-m.

Compressive Allowable, KL/r < Cc (24.7403 < 127.5514) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (24.74)² / (2 • 127.55²)) 243 /
 (5/3 + 3 • (24.74) / (8 • 127.55) - (24.74³) / (8 • 127.55³))
 = 136.9 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 8.59/136.92 + (1353.99 • 199.238/75607016)/161.73
 = 0.279

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 918735/265.214
 = 34.641 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(34.641, abs(250.0 - 34.641))
 = 215.359 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 2260.0 - cos(2024.0/2) (36.0 + 0.0 + 0.0) - 0.0
 = 200.000 mm.

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$



$$= \sqrt{(8614.9/265.214)}$$

$$= 56.994 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 200.0/56.994$$

$$= 3.509$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11448.734 / (2 \cdot 3540.0)$$

$$= 1.617 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.617 \cdot 977.517 \cdot 200.0$$

$$= 316.145 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 316.145/400023.75$$

$$= 7.750 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.5092 < 127.5514) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.51)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (3.51) / (8 \cdot 127.55) - (3.51^3) / (8 \cdot 127.55^3))$$

$$= 144.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_1/I) / S_{ba}$$

$$= 8.76/144.61 + (316.15 \cdot 215.359/8614.852) / 161.73$$

$$= 0.108$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	45.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	8.5560 cm ²
Saddle Load Q0 (Weight)	Q0	89106.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4173.2 Kg
Maximum Transverse Force	Ft	5724.4 Kg
Maximum Longitudinal Force	Fl	11448.7 Kg
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

$$= 26767.1 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 26767.1 / (8.556 \cdot 4)$$

$$= 76.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5724.4 / (8.556 \cdot 4)$$

$$= 16.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarear]:



= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $11448.73 / (82.8 \cdot 4.0)$
= 3.3900 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $2260.0 \cdot 5724.37 + 0.0$
= 12937.33 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $12937.33 / 89106.38$
= $145.19 \text{ mm.} < B_{\text{plen}}/6 \text{ --> No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[B_{\text{ltarearl}}, B_{\text{ltarears}}, B_{\text{ltareart}}]$
= $\max[0.0, 3.39, 0.0]$
= 3.3900 cm^2

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 8.427$, $c = 1.6 \text{ mm.}$, $E^* = 0.85$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.689$, Temp. Reduction = $17 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-40 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.80	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.80	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		98585.76	Kgf
Pressure used in Analysis	P	3.665	bars



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	33.12	138.00
Long. Stress at Bottom of Midspan	37.60	138.00
Long. Stress at Top of Saddles	58.52	138.00
Long. Stress at Bottom of Saddles	22.03	138.00
Tangential Shear in Shell	19.38	110.40
Circ. Stress at Horn of Saddle	162.68	172.50
Circ. Compressive Stress in Shell	6.11	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(3120.0/2 + 0) \cdot 2260.0/3540.0002$
 $= 2987.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(1919.5, 0) \cdot 2260.0/6200.0$
 $= 699.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(11448.73, 0) \cdot 2260.0/6200.0$
 $= 4173.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(11449/2 + 0) \cdot 2260.0/3540.0002$
 $= 10963.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 87622 + \max(700, 2988, 4173, 10964)$
 $= 98585.8 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(4.024 + \pi/4(4.024 \cdot 1.2)^2)$
 $= 18823.574 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100187.20	Kgf
Transverse Shear Load	5724.37	Kgf
Longitudinal Shear Load	11448.73	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3540.000 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -98586 \cdot 2150.0 [1 - (1 - 2150.0/10500.0 + (2006.8^2 - 1005.8^2) /$



$$(2 \cdot 2150.0 \cdot 10500.0) / (1 + (4 \cdot 1005.8) / (3 \cdot 10500.0)) \\ = -49938.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\ = 98586 \cdot 10500 / 4 \cdot ((1 + 2(2007^2 - 1006^2) / (10500^2)) / (1 + (4 \cdot 1006) / (3 \cdot 10500)) - 4 \cdot 2150 / 10500) \\ = 30073.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = 3.665 \cdot 2006.8 / (2 \cdot 10.4) - 30073.3 / (\pi \cdot 2006.8^2 \cdot 10.4) \\ = 33.12 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = 3.665 \cdot 2006.8 / (2 \cdot 10.4) + 30073.3 / (\pi \cdot 2006.8^2 \cdot 10.4) \\ = 37.60 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 3.665 \cdot 2006.8 / (2 \cdot 10.4) - 49938.7 / (0.1607 \cdot \pi \cdot 2006.8^2 \cdot 10.4) \\ = 58.52 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = 3.665 \cdot 2006.8 / (2 \cdot 10.4) + 49938.7 / (0.2792 \cdot \pi \cdot 2006.8^2 \cdot 10.4) \\ = 22.03 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 98586(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.8 / 3)) \\ = 51619.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 0.7988 \cdot 51619.65 / (2006.7999 \cdot 10.4) \\ = 19.38 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(2006.8 \cdot 10.4)} \\ = 112.684 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(250.0 + 1.56 \cdot \sqrt{(2006.8 \cdot 10.4)}, 2 \cdot 2150.0) \\ = 475.37 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr / S, 1) \\ = \min(137.997 / 137.997, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.6733 \cdot 98586 \cdot 0.1 / (475.369(10.4 + 1.0 \cdot 12.0)) \\ = -6.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\ = -98586 / (4(10.4 + 1.0 \cdot 12.0)475.369) - \\ 12 \cdot 0.032 \cdot 98586 \cdot 2006.8 / (10500.0(10.4 + 1.0 \cdot 12.0)^2) \\ = -162.68 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.



Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	N./mm ²
Height of Web at Center	Hw,c	200.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(2001.6 \cdot 10.4)} + 500.0)10.4^3)/12$$

$$= 6.797 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 6.797 + 75.408(136.719 - 5.2)^2$$

$$= 13050.290 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	725.1	10.4	5.2	75.4	39212.1	0.131E+05
Wearplate	500.0	12.0	16.4	60.0	98400.0	0.869E+04
Web	22.0	200.0	122.4	44.0	538559.6	0.156E+04
BasePlate	476.0	36.0	240.4	171.4	4119493.0	0.186E+05
Totals	...	...	...	350.8	4795664.5	0.419E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4795665/350.768$$

$$= 136.719 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 98585.758$$

$$= 25570.4199 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 9.1068 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2260.0 - 2001.6 \cdot \sin(150.0/2) / 1.309$$

$$= 782.993 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 20021.8652 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 64.0583 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(98586 + 1601)476.0) / (2 \cdot 3540.0 \cdot 161.733)}$$

$$= 35.004 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 3910.0679 / (5 - 1)$$

$$= 977.517 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 977.517 \cdot 476.0 / 2$$

$$= 2326.490 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 100187.203 / (3540.0 \cdot 476.0)$$

$$= 5.946 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 2326.5 \cdot 5.95$$

$$= 13832.583 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 50.16 + 107.527$$

$$= 157.687 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 13832.6 / 157.6869$$

$$= 8.603 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 22.0$$

$$= 228.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 800456 / 157.687$$

$$= 50.762 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(50.762, 250.0 - 50.762)$$

$$= 199.238 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7560.7 / 157.687)}$$

$$= 69.244 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$



$$= 2260.0 - \cos(150.0/2) (2000.0 + 12.0 + 12.0) - 36.0$$

$$= 1700.150 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(150.0/2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3540.0 - 50.8$$

$$= 3489.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (3910.068 - 3489.2) / 2$$

$$= 210.434 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lw^2 + Dd^2}$$

$$= \sqrt{1700.15^2 + 210.434^2}$$

$$= 1713.124 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 242.6)}$$

$$= 127.551$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1713.124 / 69.244$$

$$= 24.740$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 11448.7 / (2 \cdot 3540.0) \cdot 977.517 \cdot 1713.12 / 2$$

$$= 1353.991 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($24.7403 < 127.5514$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.74)^2 / (2 \cdot 127.55^2)) 243 /$$

$$(5/3 + 3 \cdot (24.74) / (8 \cdot 127.55) - (24.74^3) / (8 \cdot 127.55^3))$$

$$= 136.9 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.6 / 136.92 + (1353.99 \cdot 199.238 / 75607016) / 161.73$$

$$= 0.279$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 918735 / 265.214$$

$$= 34.641 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(34.641, \text{abs}(250.0 - 34.641))$$

$$= 215.359 \text{ mm.}$$

Length of Inner Rib [L]:



= Saddle Height - Outside Radius - Bpchk
 = $2260.0 - \cos(2024.0/2) (36.0 + 0.0 + 0.0) - 0.0$
 = 200.000 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{8614.9/265.214}$
 = 56.994 mm.

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 200.0/56.994$
 = 3.509

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = $11448.734 / (2 \cdot 3540.0)$
 = 1.617 Kg/mm.

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = $1.617 \cdot 977.517 \cdot 200.0$
 = 316.145 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $316.145/400023.75$
 = 7.750 N./mm²

Compressive Allowable, KL/r < Cc (3.5092 < 127.5514) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (3.51)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (3.51) / (8 \cdot 127.55) - (3.51^3) / (8 \cdot 127.55^3))$
 = 144.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = $8.77/144.61 + (316.15 \cdot 215.359/8614.852)/161.73$
 = 0.109

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	45.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	8.5560 cm ²
Saddle Load Q0 (Weight)	Q0	89223.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	4173.2 Kg
Maximum Transverse Force	Ft	5724.4 Kg
Maximum Longitudinal Force	Fl	11448.7 Kg
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = FI / (Bolt Area • Number of Bolts)
 = $11448.7 / (8.556 \cdot 4)$
 = 32.8 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = $5724.4 / (8.556 \cdot 4)$
 = 16.4 N./mm². Must be less than 82.8 N./mm².



Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 11448.73/(82.8 • 4.0)
= 3.3900 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 2260.0 • 5724.37 + 0.0
= 12937.33 Kg-m.

Eccentricity (e):
= Rmom / QO
= 12937.33/89223.58
= 145.00 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 3.39, 0.0]
= 3.3900 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 8.427, c = 1.6 mm., E* = 0.85
Thickness Ratio = tr • E*/(tg - c) = 0.689, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

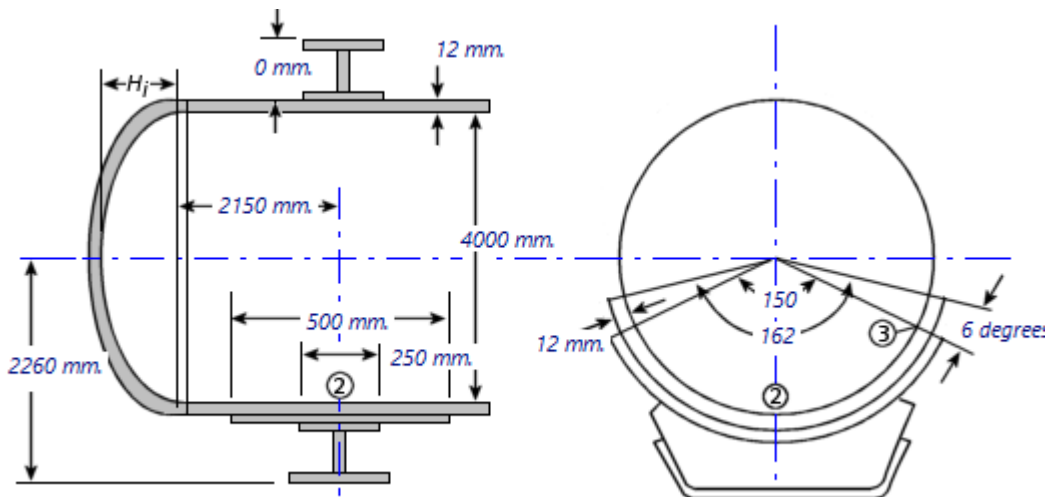
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**ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:**

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.00	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.00	

**Results for Horizontal Vessel Analysis:**

Saddle Force Q, Test Case, no Ext. Forces	89498.93	Kgf
Pressure used in Analysis	P 4.747	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	37.91	234.00
Long. Stress at Bottom	of Midspan	41.44	234.00
Long. Stress at Top	of Saddles	57.91	234.00
Long. Stress at Bottom	of Saddles	29.18	234.00
Tangential Shear in Shell		15.25	138.58
Circ. Stress at Horn of Saddle		129.23	350.99
Circ. Compressive Stress in Shell		5.00	234.00

**Intermediate Results: Saddle Reaction Q due to Wind or Seismic:**

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (1560.0/2 + 0) \cdot 2260.0/3540.0002$$

$$= 1493.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(959.75, 0) \cdot 2260.0/6200.0$$

$$= 349.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 88005 + \max(350, 1494, 0, 0)$$

$$= 89498.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (4.024 + \pi/4 (4.024 \cdot 1.2)^2)$$

$$= 18823.574 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91100.37	Kgf
Transverse Shear Load	780.00	Kgf
Longitudinal Shear Load	959.75	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3540.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

*The suffix 'p' denotes the values for a wear plate if it exists.***Note:** Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -89499 \cdot 2150.0 [1 - (1 - 2150.0/10500.0 + (2006.0^2 - 1005.0^2) / (2 \cdot 2150.0 \cdot 10500.0)) / (1 + (4 \cdot 1005.0) / (3 \cdot 10500.0))]$$

$$= -45328.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 89499 \cdot 10500 / 4 ((1 + 2(2006^2 - 1005^2)/(10500^2)) / (1 + (4 \cdot 1005) / (3 \cdot 10500)) - 4 \cdot 2150 / 10500)$$

$$= 27315.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) - 27315.2 / (\pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 37.91 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) + 27315.2 / (\pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 41.44 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) - 45328.6 / (0.1607 \cdot \pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 57.91 \text{ N./mm}^2$$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) + 45328.6 / (0.2792 \cdot \pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 29.18 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 89499(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.0 / 3))$$

$$= 46866.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 46866.0 / (2006.0 \cdot 12.0)$$

$$= 15.25 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2006.0 \cdot 12.0)}$$

$$= 121.018 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(2006.0 \cdot 12.0)}, 2 \cdot 2150.0)$$

$$= 492.04 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 89499 \cdot 0.1 / (492.036(12.0 + 1.0 \cdot 12.0))$$

$$= -5.00 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -89499 / (4(12.0 + 1.0 \cdot 12.0)492.036) -$$

$$12 \cdot 0.032 \cdot 89499 \cdot 2006.0 / (10500.0(12.0 + 1.0 \cdot 12.0)^2)$$

$$= -129.23 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.
Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	200.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2000.0 \cdot 12.0)} + 500.0)12.0^3) / 12$$

$$= 10.680 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 10.68 + 89.001(133.217 - 6.0)^2$$

$$= 14414.747 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	741.7	12.0	6.0	89.0	53400.5	0.144E+05
Wearplate	500.0	12.0	18.0	60.0	108000.0	0.797E+04



Web	22.0	200.0	124.0	44.0	545599.6	0.150E+04
BasePlate	476.0	36.0	242.0	171.4	4146910.5	0.205E+05
Totals	...	...	...	364.4	4853910.5	0.444E+05

Distance to Centroid [C1]:

= AY / A

= 4853911/364.361

= 133.217 mm.

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 150.0/2

= 105.0

Saddle Splitting Coefficient [K1]:

= $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$ = $(1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$

= 0.2594

Saddle Splitting Force [Fh]:

= $K1 \cdot Q$

= 0.259 • 89498.93

= 23213.5488 Kgf

Tension Stress, $St = (Fh/As)$ = 8.2674 N./mm²Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str}$ = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= $B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$

= 2260.0 - 2000.0 • sin(150.0/2)/1.309

= 784.174 mm.

Bending Moment, $M = Fh \cdot d$

= 18203.8203 Kg-m.

Bending Stress, $Sb = (M \cdot C1 / I)$ = 53.6176 N./mm²Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str}$ = 173.3302 N./mm²**Minimum Thickness of Baseplate per Moss:**= $\sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$ = $\sqrt{3(89499 + 1601)476.0 / (2 \cdot 3540.0 \cdot 173.33)}$

= 32.243 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$ = $2 \cdot \cos(90 - 150.0/2) (2000.0 + 12.0 + 12.0)$

= 3910.068 mm.

Distance between Ribs [e]:

= $\text{Web Length} / (N_{\text{ribs}} - 1)$

= 3910.0679/(5 - 1)

= 977.517 mm.

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= 977.517 • 476.0/2

= 2326.490 cm²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= 91100.367/(3540.0 • 476.0)

= 5.406 Kgf/cm²

Axial Load [P]:

= $Ap \cdot Bp$

= 2326.5 • 5.41

= 12577.988 Kgf



Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 50.16 + 107.527 \\
 &= 157.687 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 12578.0/157.6869 \\
 &= 7.822 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250.0 - 22.0 \\
 &= 228.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 800456/157.687 \\
 &= 50.762 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(\text{ytot}, \text{Saddle Width} - \text{ytot}) \\
 &= \max(50.762, 250.0 - 50.762) \\
 &= 199.238 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(7560.7/157.687)} \\
 &= 69.244 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2260.0 - \cos(150.0/2)(2000.0 + 12.0 + 12.0) - 36.0 \\
 &= 1700.150 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(150.0/2)(2000.0 + 12.0 + 12.0) \\
 &= 3910.068 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 3540.0 - 50.8 \\
 &= 3489.200 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd)/2 \\
 &= (3910.068 - 3489.2)/2 \\
 &= 210.434 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{(Lv^2 + Dd^2)} \\
 &= \sqrt{(1700.15^2 + 210.434^2)} \\
 &= 1713.124 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)} \\
 &= 123.211
 \end{aligned}$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1713.124/69.244$$

$$= 24.740$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 959.7 / (2 \cdot 3540.0) \cdot 977.517 \cdot 1713.12 / 2$$

$$= 113.505 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (24.7403 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (24.74)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (24.74) / (8 \cdot 123.21) - (24.74^3) / (8 \cdot 123.21^3))$$

$$= 146.3 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.82 / 146.33 + (113.51 \cdot 199.238 / 75607016) / 173.33$$

$$= 0.070$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 918735 / 265.214$$

$$= 34.641 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(34.641, \text{abs}(250.0 - 34.641))$$

$$= 215.359 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2260.0 - \cos(2024.0 / 2) (36.0 + 0.0 + 0.0) - 0.0$$

$$= 200.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (8614.9 / 265.214) }$$

$$= 56.994 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 200.0 / 56.994$$

$$= 3.509$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 959.75 / (2 \cdot 3540.0)$$

$$= 0.136 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.136 \cdot 977.517 \cdot 200.0$$

$$= 26.503 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 26.503 / 400023.75$$



$$= 0.650 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.5092 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.51)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.51) / (8 \cdot 123.21) - (3.51^3) / (8 \cdot 123.21^3))$$

$$= 154.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 7.97 / 154.94 + (26.5 \cdot 215.359 / 8614.852) / 173.33$$

$$= 0.055$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	45.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	8.5560 cm ²
Saddle Load Q0 (Weight)	Q0	89606.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	349.8 Kg
Maximum Transverse Force	Ft	780.0 Kg
Maximum Longitudinal Force	Fl	959.7 Kg
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 959.7 / (8.556 \cdot 4)$$

$$= 2.8 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 780.0 / (8.556 \cdot 4)$$

$$= 2.2 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 959.75 / (68.4 \cdot 4.0)$$

$$= 0.3440 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2260.0 \cdot 780.0 + 0.0$$

$$= 1762.84 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 1762.84 / 89606.47$$

$$= 19.67 \text{ mm.} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.344, 0.0]$$



= 0.3440 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.00	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		89616.12	Kgf
Pressure used in Analysis	P	4.747	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	37.91	234.00
Long. Stress at Bottom of Midspan	41.44	234.00
Long. Stress at Top of Saddles	57.94	234.00
Long. Stress at Bottom of Saddles	29.17	234.00
Tangential Shear in Shell	15.27	138.58
Circ. Stress at Horn of Saddle	129.40	350.99
Circ. Compressive Stress in Shell	5.01	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (1560.0/2 + 0) \cdot 2260.0/3540.0002$$

$$= 1493.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(959.75, 0) \cdot 2260.0/6200.0$$

$$= 349.8 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 88122 + \max(350, 1494, 0, 0)$$

$$= 89616.1 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (4.024 + \pi/4 (4.024 \cdot 1.2)^2)$$

$$= 18823.574 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

Vertical Load (including saddle weight)	91217.57	Kgf
Transverse Shear Load	780.00	Kgf
Longitudinal Shear Load	959.75	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 3540.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h_m)}{3L} \right) \right]$$

$$= -89616 \cdot 2150.0 \left[1 - \left(1 - \frac{2150.0}{10500.0} + \frac{(2006.0^2 - 1005.0^2)}{(2 \cdot 2150.0 \cdot 10500.0)} \right) / \left(1 + \frac{(4 \cdot 1005.0)}{3 \cdot 10500.0} \right) \right]$$

$$= -45387.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{1 + 2(R_m^2 - h_m^2)/(L^2)}{1 + (4h_m^2)/(3L)} - \frac{4a}{L} \right)$$

$$= 89616 \cdot 10500 / 4 \left(\frac{1 + 2(2006^2 - 1005^2)/(10500^2)}{1 + (4 \cdot 1005)/(3 \cdot 10500)} - \frac{4 \cdot 2150}{10500} \right)$$

$$= 27350.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) - 27350.9 / (\pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 37.91 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) + 27350.9 / (\pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 41.44 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) - 45387.9 / (0.1607 \cdot \pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 57.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4.747 \cdot 2006.0 / (2 \cdot 12.0) + 45387.9 / (0.2792 \cdot \pi \cdot 2006.0^2 \cdot 12.0)$$

$$= 29.17 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 89616(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.0 / 3))$$

$$= 46927.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 46927.37 / (2006.0 \cdot 12.0)$$

$$= 15.27 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2006.0 \cdot 12.0)}$$

$$= 121.018 \text{ mm.}$$



Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{ 2006.0 \cdot 12.0 }, 2 \cdot 2150.0)$$

$$= 492.04 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 89616 \cdot 0.1 / (492.036(12.0 + 1.0 \cdot 12.0))$$

$$= -5.01 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -89616 / (4(12.0 + 1.0 \cdot 12.0)492.036) -$$

$$12 \cdot 0.032 \cdot 89616 \cdot 2006.0 / (10500.0(12.0 + 1.0 \cdot 12.0)^2)$$

$$= -129.40 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.
Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	200.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 2000.0 \cdot 12.0 } + 500.0) 12.0^3) / 12$$

$$= 10.680 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 10.68 + 89.001(133.217 - 6.0)^2$$

$$= 14414.747 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	741.7	12.0	6.0	89.0	53400.5	0.144E+05
Wearplate	500.0	12.0	18.0	60.0	108000.0	0.797E+04
Web	22.0	200.0	124.0	44.0	545599.6	0.150E+04
BasePlate	476.0	36.0	242.0	171.4	4146910.5	0.205E+05
Totals	...	...	...	364.4	4853910.5	0.444E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4853911 / 364.361$$

$$= 133.217 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:



$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.259 \cdot 89616.125 \\
 &= 23243.9473 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 8.2782 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 2260.0 - 2000.0 \cdot \sin(150.0/2) / 1.309 \\
 &= 784.174 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 18227.6602 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 53.6878 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\
 &= \sqrt{(3 \cdot (89616 + 1601) \cdot 476.0 / (2 \cdot 3540.0 \cdot 173.33))} \\
 &= 32.264 \text{ mm.}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 150.0/2) \cdot (2000.0 + 12.0 + 12.0) \\
 &= 3910.068 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 3910.0679 / (5 - 1) \\
 &= 977.517 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 977.517 \cdot 476.0 / 2 \\
 &= 2326.490 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 91217.578 / (3540.0 \cdot 476.0) \\
 &= 5.413 \text{ Kg/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 2326.5 \cdot 5.41 \\
 &= 12594.170 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 50.16 + 107.527 \\
 &= 157.687 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Compressive Stress [Sc]:} \\
 &= P/Ar \\
 &= 12594.2 / 157.6869 \\
 &= 7.833 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04



Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250.0 - 22.0 \\
 &= 228.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 800456 / 157.687 \\
 &= 50.762 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(y_{tot}, \text{Saddle Width} - y_{tot}) \\
 &= \max(50.762, 250.0 - 50.762) \\
 &= 199.238 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\
 &= \sqrt{ (7560.7 / 157.687) } \\
 &= 69.244 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2260.0 - \cos(150.0 / 2) (2000.0 + 12.0 + 12.0) - 36.0 \\
 &= 1700.150 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(150.0 / 2) (2000.0 + 12.0 + 12.0) \\
 &= 3910.068 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 3540.0 - 50.8 \\
 &= 3489.200 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (3910.068 - 3489.2) / 2 \\
 &= 210.434 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{ (Lv^2 + Dd^2) } \\
 &= \sqrt{ (1700.15^2 + 210.434^2) } \\
 &= 1713.124 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) } \\
 &= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994 \text{E}+09 / 260.0) } \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL / r \\
 &= 1 \cdot 1713.124 / 69.244 \\
 &= 24.740
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\
 &= 959.7 / (2 \cdot 3540.0) \cdot 977.517 \cdot 1713.12 / 2 \\
 &= 113.505 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($24.7403 < 123.2105$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (24.74)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (24.74) / (8 \cdot 123.21) - (24.74^3) / (8 \cdot 123.21^3)) \\
 &= 146.3 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 7.83 / 146.33 + (113.51 \cdot 199.238 / 75607016) / 173.33
 \end{aligned}$$



= 0.070

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 918735/265.214

= 34.641 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(34.641, abs(250.0 - 34.641))

= 215.359 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 2260.0 - cos(2024.0/2)(36.0 + 0.0 + 0.0) - 0.0

= 200.000 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{8614.9/265.214}$

= 56.994 mm.

Slenderness ratio [KL/r]:

= KL/r

= 1 • 200.0/56.994

= 3.509

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 959.75/(2 • 3540.0)

= 0.136 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.136 • 977.517 • 200.0

= 26.503 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 26.503/400023.75

= 0.650 N./mm²

Compressive Allowable, KL/r < Cc (3.5092 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (3.51)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (3.51) / (8 \cdot 123.21) - (3.51^3) / (8 \cdot 123.21^3))$

= 154.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I) / Sba

= 7.99/154.94 + (26.5 • 215.359/8614.852) / 173.33

= 0.055

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.



Distance from Bolts to Edge	Edgedis	45.0 mm.
Nominal Bolt Diameter	Bnd	42.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	8.5560 cm ²
Saddle Load Q0 (Weight)	Q0	89723.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	349.8 Kgf
Maximum Transverse Force	Ft	780.0 Kgf
Maximum Longitudinal Force	Fl	959.7 Kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 959.7 / (8.556 \cdot 4)$$

$$= 2.8 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 780.0 / (8.556 \cdot 4)$$

$$= 2.2 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 959.75 / (68.4 \cdot 4.0)$$

$$= 0.3440 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2260.0 \cdot 780.0 + 0.0$$

$$= 1762.84 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 1762.84 / 89723.66$$

$$= 19.65 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.344, 0.0]$$

$$= 0.3440 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

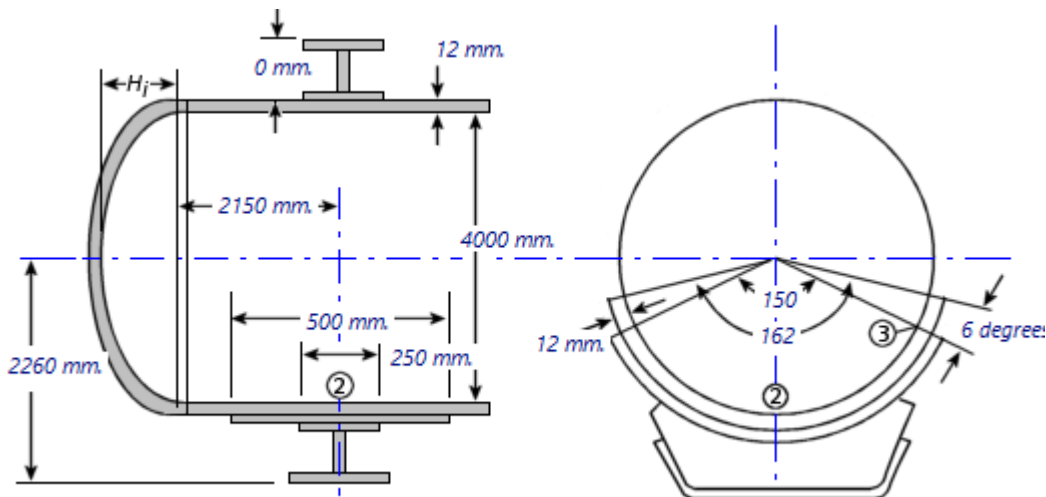
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.80	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.80	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	98468.55	Kgf
Pressure used in Analysis	P -0.517	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-7.23	-77.43
Long. Stress at Top of Midspan	7.23	138.00
Long. Stress at Bottom of Midspan	-2.75	-77.43
Long. Stress at Bottom of Midspan	2.75	138.00
Long. Stress at Top of Saddles	18.15	138.00
Long. Stress at Bottom of Saddles	-18.30	-77.43
Long. Stress at Bottom of Saddles	18.30	138.00
Tangential Shear in Shell	19.35	110.40
Circ. Stress at Horn of Saddle	162.48	172.50



Circ. Compressive Stress in Shell

6.11

138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3120.0/2 + 0) \cdot 2260.0/3540.0002$$

$$= 2987.8 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(1919.5, 0) \cdot 2260.0/6200.0$$

$$= 699.7 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(11448.73, 0) \cdot 2260.0/6200.0$$

$$= 4173.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(11449/2 + 0) \cdot 2260.0/3540.0002$$

$$= 10963.6 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 87505 + \max(700, 2988, 4173, 10964)$$

$$= 98468.6 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(4.024 + \pi/4(4.024 \cdot 1.2)^2)$$

$$= 18823.574 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100070.00	Kgf
Transverse Shear Load	5724.37	Kgf
Longitudinal Shear Load	11448.73	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3540.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -98469 \cdot 2150.0 [1 - (1 - 2150.0/10500.0 + (2006.8^2 - 1005.8^2) / (2 \cdot 2150.0 \cdot 10500.0)) / (1 + (4 \cdot 1005.8) / (3 \cdot 10500.0))]$$

$$= -49879.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 98469 \cdot 10500 / 4 ((1 + 2(2007^2 - 1006^2)/(10500^2)) / (1 + (4 \cdot 1006) / (3 \cdot 10500)) - 4 \cdot 2150 / 10500)$$

$$= 30037.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:



$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) - 30037.6 / (\pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -7.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) + 30037.6 / (\pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -2.75 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) - 49879.3 / (0.1607 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= 18.15 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) + 49879.3 / (0.2792 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -18.30 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 98469(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.8/3))$$

$$= 51558.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.7988 \cdot 51558.28 / (2006.7999 \cdot 10.4)$$

$$= 19.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2006.8 \cdot 10.4)}$$

$$= 112.684 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(2006.8 \cdot 10.4)}, 2 \cdot 2150.0)$$

$$= 475.37 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 98469 \cdot 0.1 / (475.369(10.4 + 1.0 \cdot 12.0))$$

$$= -6.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -98469 / (4(10.4 + 1.0 \cdot 12.0)475.369) -$$

$$12 \cdot 0.032 \cdot 98469 \cdot 2006.8 / (10500.0(10.4 + 1.0 \cdot 12.0)^2)$$

$$= -162.48 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.
Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	N./mm ²
Height of Web at Center	Hw,c	200.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(2001.6 \cdot 10.4)} + 500.0)^3) / 12$$

$$= 6.797 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 6.797 + 75.408 (136.719 - 5.2)^2$$

$$= 13050.290 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	725.1	10.4	5.2	75.4	39212.1	0.131E+05
Wearplate	500.0	12.0	16.4	60.0	98400.0	0.869E+04
Web	22.0	200.0	122.4	44.0	538559.6	0.156E+04
BasePlate	476.0	36.0	240.4	171.4	4119493.0	0.186E+05
Totals	...	...	...	350.8	4795664.5	0.419E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4795665 / 350.768$$

$$= 136.719 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 98468.555$$

$$= 25540.0215 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.0960 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2260.0 - 2001.6 \cdot \sin(150.0/2) / 1.309$$

$$= 782.993 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19998.0645 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 63.9821 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3 \cdot (98469 + 1601) \cdot 476.0) / (2 \cdot 3540.0 \cdot 161.733)}$$

$$= 34.984 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3910.0679 / (5 - 1)$$

$$= 977.517 \text{ mm.}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 977.517 \cdot 476.0 / 2$$

$$= 2326.490 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 100070.008 / (3540.0 \cdot 476.0)$$

$$= 5.939 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2326.5 \cdot 5.94$$

$$= 13816.401 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 50.16 + 107.527$$

$$= 157.687 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 13816.4 / 157.6869$$

$$= 8.593 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 22.0$$

$$= 228.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 800456 / 157.687$$

$$= 50.762 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(50.762, 250.0 - 50.762)$$

$$= 199.238 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 7560.7 / 157.687 }$$

$$= 69.244 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2260.0 - \cos(150.0 / 2) (2000.0 + 12.0 + 12.0) - 36.0$$

$$= 1700.150 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(150.0 / 2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3540.0 - 50.8$$



$$= 3489.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (L_w - B_d) / 2 \\ &= (3910.068 - 3489.2) / 2 \\ &= 210.434 \text{ mm.} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{(L_v^2 + D_d^2)} \\ &= \sqrt{(1700.15^2 + 210.434^2)} \\ &= 1713.124 \text{ mm.} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 242.6)} \\ &= 127.551 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 1713.124 / 69.244 \\ &= 24.740 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\ &= 11448.7 / (2 \cdot 3540.0) \cdot 977.517 \cdot 1713.12 / 2 \\ &= 1353.991 \text{ Kg-m.} \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($24.7403 < 127.5514$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (24.74)^2 / (2 \cdot 127.55^2)) 243 / \\ &\quad (5/3 + 3 \cdot (24.74) / (8 \cdot 127.55) - (24.74^3) / (8 \cdot 127.55^3)) \\ &= 136.9 \text{ N/mm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\ &= 8.59 / 136.92 + (1353.99 \cdot 199.238 / 75607016) / 161.73 \\ &= 0.279 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 918735 / 265.214 \\ &= 34.641 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot})) \\ &= \max(34.641, \text{abs}(250.0 - 34.641)) \\ &= 215.359 \text{ mm.} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \text{Outside Radius} - B_{pthk} \\ &= 2260.0 - \cos(2024.0/2) (36.0 + 0.0 + 0.0) - 0.0 \\ &= 200.000 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(8614.9 / 265.214)} \\ &= 56.994 \text{ mm.} \end{aligned}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



$$= 1 \cdot 200.0/56.994$$

$$= 3.509$$

Unit Force [Force,u]:

$$= F1 / (2 \cdot \text{Baseplate Length})$$

$$= 11448.734 / (2 \cdot 3540.0)$$

$$= 1.617 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.617 \cdot 977.517 \cdot 200.0$$

$$= 316.145 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 316.145/400023.75$$

$$= 7.750 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.5092 < 127.5514) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.51)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (3.51) / (8 \cdot 127.55) - (3.51)^3 / (8 \cdot 127.55^3))$$

$$= 144.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 8.76/144.61 + (316.15 \cdot 215.359/8614.852) / 161.73$$

$$= 0.108$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	2006.80	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	10500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2150.00	mm.
Saddle Width	b1 or b	250.00	mm.
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	500.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1005.80	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2260.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		98585.76	Kgf
Pressure used in Analysis	P	-0.517	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-7.23	-77.43
Long. Stress at Top	of Midspan	7.23	138.00
Long. Stress at Bottom	of Midspan	-2.75	-77.43
Long. Stress at Bottom	of Midspan	2.75	138.00
Long. Stress at Top	of Saddles	18.18	138.00



Long. Stress at Bottom of Saddles	-18.32	-77.43
Long. Stress at Bottom of Saddles	18.32	138.00
Tangential Shear in Shell	19.38	110.40
Circ. Stress at Horn of Saddle	162.68	172.50
Circ. Compressive Stress in Shell	6.11	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3120.0/2 + 0) \cdot 2260.0/3540.0002$$

$$= 2987.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1919.5, 0) \cdot 2260.0/6200.0$$

$$= 699.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(11448.73, 0) \cdot 2260.0/6200.0$$

$$= 4173.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(11449/2 + 0) \cdot 2260.0/3540.0002$$

$$= 10963.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 87622 + \max(700, 2988, 4173, 10964)$$

$$= 98585.8 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(4.024 + \pi/4(4.024 \cdot 1.2)^2)$$

$$= 18823.574 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100187.20	Kgf
Transverse Shear Load	5724.37	Kgf
Longitudinal Shear Load	11448.73	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3540.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -98586 \cdot 2150.0 [1 - (1 - 2150.0/10500.0 + (2006.8^2 - 1005.8^2)/(2 \cdot 2150.0 \cdot 10500.0))/(1 + (4 \cdot 1005.8)/(3 \cdot 10500.0))]$$

$$= -49938.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$



$$= 98586 \cdot 10500 / 4 \cdot ((1 + 2(2007^2 - 1006^2) / (10500^2)) / (1 + (4 \cdot 1006) / (3 \cdot 10500)) - 4 \cdot 2150 / 10500)$$

$$= 30073.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) - 30073.3 / (\pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -7.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) + 30073.3 / (\pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -2.75 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) - 49938.7 / (0.1607 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= 18.18 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.517 \cdot 2006.8 / (2 \cdot 10.4) + 49938.7 / (0.2792 \cdot \pi \cdot 2006.8^2 \cdot 10.4)$$

$$= -18.32 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 98586(10500.0 - 2 \cdot 2150.0) / (10500.0 + (4 \cdot 1005.8 / 3))$$

$$= 51619.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.7988 \cdot 51619.65 / (2006.7999 \cdot 10.4)$$

$$= 19.38 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(2006.8 \cdot 10.4)}$$

$$= 112.684 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(2006.8 \cdot 10.4)}, 2 \cdot 2150.0)$$

$$= 475.37 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(137.997 / 137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.6733 \cdot 98586 \cdot 0.1 / (475.369(10.4 + 1.0 \cdot 12.0))$$

$$= -6.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -98586 / (4(10.4 + 1.0 \cdot 12.0) 475.369) -$$

$$12 \cdot 0.032 \cdot 98586 \cdot 2006.8 / (10500.0(10.4 + 1.0 \cdot 12.0)^2)$$

$$= -162.68 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3540.0002	mm.
Baseplate Thickness	Bpthk	36.0000	mm.
Baseplate Width	Bpwid	476.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	22.0000	mm.
Web Thickness	Webtk	22.0000	mm.
Web Location	Webloc	Side	



Saddle Design Temperature 21.1 °C
 Saddle Yield Stress Sy 242.6 N./mm²
 Height of Web at Center Hw,c 200.0 mm.
 Friction Coefficient mu 0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(2001.6 \cdot 10.4)} + 500.0)^3) / 12$
 $= 6.797 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 6.797 + 75.408 (136.719 - 5.2)^2$
 $= 13050.290 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	725.1	10.4	5.2	75.4	39212.1	0.131E+05
Wearplate	500.0	12.0	16.4	60.0	98400.0	0.869E+04
Web	22.0	200.0	122.4	44.0	538559.6	0.156E+04
BasePlate	476.0	36.0	240.4	171.4	4119493.0	0.186E+05
Totals	...	...	...	350.8	4795664.5	0.419E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 4795665 / 350.768$
 $= 136.719 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 150.0 / 2$
 $= 105.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$
 $= 0.2594$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.259 \cdot 98585.758$
 $= 25570.4199 \text{ Kg}$

Tension Stress, St = (Fh/As) = 9.1068 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 145.5600 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 2260.0 - 2001.6 \cdot \sin(150.0/2) / 1.309$
 $= 782.993 \text{ mm.}$

Bending Moment, M = Fh * d = 20021.8652 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 64.0583 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 161.7334 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(98586 + 1601) \cdot 476.0) / (2 \cdot 3540.0 \cdot 161.733))}$
 $= 35.004 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 150.0/2) (2000.0 + 12.0 + 12.0)$



= 3910.068 mm.

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 3910.0679 / (5 - 1)

= 977.517 mm.

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 977.517 • 476.0 / 2

= 2326.490 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 100187.203 / (3540.0 • 476.0)

= 5.946 Kg/cm²

Axial Load [P]:

= Ap • Bp

= 2326.5 • 5.95

= 13832.583 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 50.16 + 107.527

= 157.687 cm²

Compressive Stress [Sc]:

= P / Ar

= 13832.6 / 157.6869

= 8.603 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.582E+04
Web	488.8	22.0	11.0	107.5	118279.5	0.174E+04
Totals	...	...	...	157.7	800455.5	0.756E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 250.0 - 22.0

= 228.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A

= 800456 / 157.687

= 50.762 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)

= max(50.762, 250.0 - 50.762)

= 199.238 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{7560.7 / 157.687}$

= 69.244 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2) (radius + shlthk + wpdthk) - bpthk

= 2260.0 - cos(150.0 / 2) (2000.0 + 12.0 + 12.0) - 36.0

= 1700.150 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2) (radius + shlthk + wpdthk)



$$= 2 \cdot \sin(150.0/2) (2000.0 + 12.0 + 12.0)$$

$$= 3910.068 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3540.0 - 50.8$$

$$= 3489.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (3910.068 - 3489.2) / 2$$

$$= 210.434 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1700.15^2 + 210.434^2)}$$

$$= 1713.124 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 242.6)}$$

$$= 127.551$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1713.124 / 69.244$$

$$= 24.740$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 11448.7 / (2 \cdot 3540.0) \cdot 977.517 \cdot 1713.12 / 2$$

$$= 1353.991 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($24.7403 < 127.5514$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (24.74)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (24.74) / (8 \cdot 127.55) - (24.74^3) / (8 \cdot 127.55^3))$$

$$= 136.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.6 / 136.92 + (1353.99 \cdot 199.238 / 75607016) / 161.73$$

$$= 0.279$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	228.0	136.0	50.2	682176.0	0.733E+04
Web	977.5	22.0	11.0	215.1	236559.1	0.129E+04
Totals	...	...	...	265.2	918735.1	0.861E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 918735 / 265.214$$

$$= 34.641 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(34.641, \text{abs}(250.0 - 34.641))$$

$$= 215.359 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 2260.0 - \cos(2024.0/2) (36.0 + 0.0 + 0.0) - 0.0$$

$$= 200.000 \text{ mm.}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{8614.9 / 265.214}$$

$$= 56.994 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 200.0 / 56.994$$

$$= 3.509$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11448.734 / (2 \cdot 3540.0)$$

$$= 1.617 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.617 \cdot 977.517 \cdot 200.0$$

$$= 316.145 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 316.145 / 400023.75$$

$$= 7.750 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.5092 < 127.5514) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.51)^2 / (2 \cdot 127.55^2)) 243 / (5/3 + 3 \cdot (3.51) / (8 \cdot 127.55) - (3.51^3) / (8 \cdot 127.55^3))$$

$$= 144.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 8.77 / 144.61 + (316.15 \cdot 215.359 / 8614.852) / 161.73$$

$$= 0.109$$

*Review notes about nozzle loadings and supports in the Warnings report.***PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.**



Input, Nozzle Desc: M1-24"

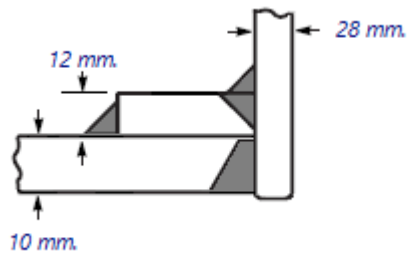
From: 10

Pressure for Reinforcement Calculations	P	4.731	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.73 \cdot 4003.2 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.73)$
 $= 6.1742 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.73 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.73)$
 $= 1.0437 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0288 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm.
Parallel to Vessel Wall, opening length	d	556.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		21.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

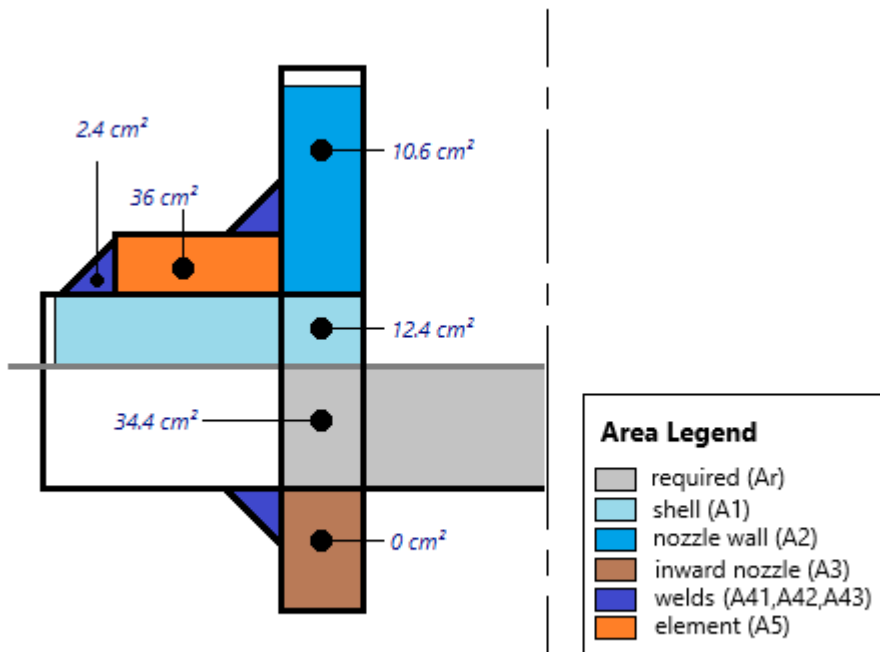


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	34.378	20.490	NA
Area in Shell	A1	12.393	5.791	NA
Area in Nozzle Wall	A2	10.650	10.656	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.430	2.430	NA
Area in Element	A5	36.000	36.000	NA
TOTAL AREA AVAILABLE	Atot	61.473	54.877	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	683.8120	12.0000 mm.
Based on given Pad Diameter:	909.6000	2.9685 mm.
Based on Shell or Nozzle Thickness:	698.6544	10.0000 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (556.8 \cdot 1.0 \cdot 8.4 - 1.0 \cdot 2 \cdot 26.4 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 34.378 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 556.8(1.0 \cdot 8.4 - 1.0 \cdot 6.174) - 2 \cdot 26.4$$

$$(1.0 \cdot 8.4 - 1.0 \cdot 6.1742) \cdot (1 - 1.0)$$

$$= 12.393 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 21.0) \cdot (26.4 - 1.04) \cdot 1.0$$

$$= 10.650 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + W_p^2 \cdot \text{fr4}$$

$$= (0.99) \cdot 1.0 + (0.0) \cdot 1.0 + 12.0^2 \cdot 1.0$$

$$= 2.430 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr4}$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 36.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.6437 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.6437 \text{ mm}.$$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 28.0$, $t_r = 1.044$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.04$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.174$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.174$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4.73/19.60 = 0.241

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 10.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	5.2000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (34.3782 - 12.393 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 8.4 - 6.1742)) \cdot 138)$
 $= 32590.19 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (10.6497 + 36.0 + 2.43 - 0.0 \cdot 1.0) \cdot 138$
 $= 69062.80 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (10.6497 + 0.0 + 1. + (4.4352)) \cdot 138$
 $= 22633.93 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (10.6497 + 0.0 + 2.43 + 36.0 + (4.4352)) \cdot 138$
 $= 75303.81 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 118220. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 238226. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 119654. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (10.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 83758. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (118220 + 238226) = 356446 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (66025 + 119654 + 83758 + 0) = 269438 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (118220 + 83758 + 0) = 201978 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 356446 Kgf, must exceed W = 32590 Kgf or W1 = 69062 Kgf

Path 2-2 = 269437 Kgf, must exceed W = 32590 Kgf or W2 = 22633 Kgf

Path 3-3 = 201977 Kgf, must exceed W = 32590 Kgf or W3 = 75303 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 12.8977 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.8978 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.814 %

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Input, Nozzle Desc: K2B-2"

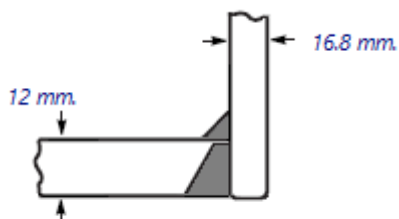
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Pressure for Reinforcement Calculations	P	4.925	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	1646.3336	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.93 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.93)$
= 7.1597 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.93}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0965 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

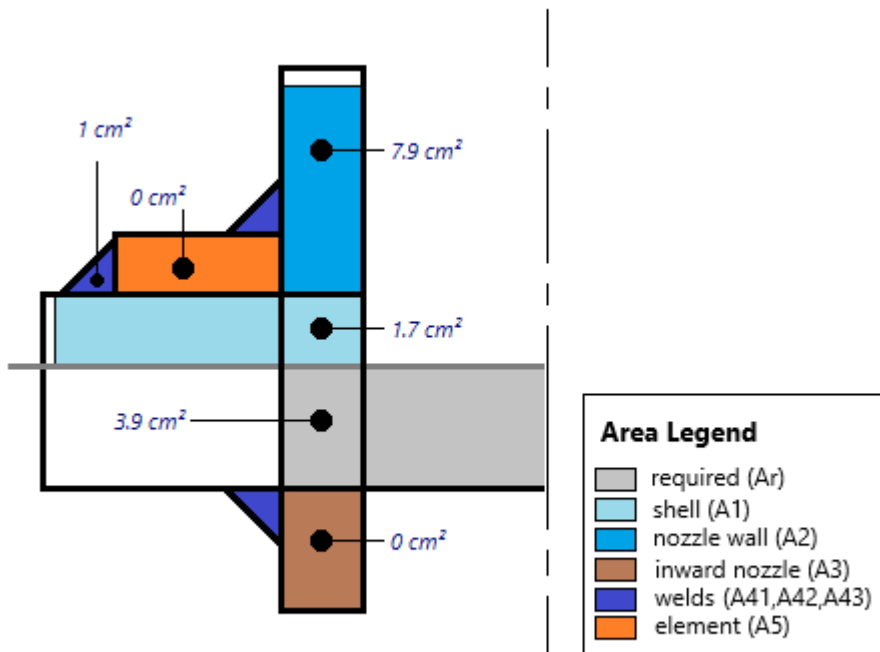


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.866	1.832	NA
Area in Shell	A1	1.750	1.952	NA
Area in Nozzle Wall	A2	7.854	7.749	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.604	10.702	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 7.1597 \cdot 1.0 + 2 \cdot 15.2 \cdot 7.1597 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.866 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 7.16) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 7.1597) \cdot (1 - 1.0)$$

$$= 1.750 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.1)1.0$$

$$= 7.854 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.8975 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.7597 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.7597 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb_2 = 2.3503 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 6.4000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb_3, \max(tb_1, tb_2)] \\
 &= \min[6.4, \max(8.7597, 4.1)] \\
 &= 6.4000 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(t_a, t_b) \\
 &= \max(1.8975, 6.4) \\
 &= 6.4000 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)**

Govrn. thk, $t_g = 12.0$, $t_r = 7.16$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.688$, Temp. Reduction = $17 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-3 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-20 \text{ }^\circ\text{C}$
 Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.93/51.00 = 0.097$$

Weld Size Calculations, Description: K2B-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 10.4000 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot S_v) \\
 &= \max(0, (3.8662 - 1.7498 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 7.1597)) \cdot 138) \\
 &= 4364.35 \text{ Kg}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\
 &= (7.8538 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 12458.68 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$



$$= (7.8538 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16907.55 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (7.8538 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16907.55 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 4364 Kgf	or W1 = 12458 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 4364 Kgf	or W2 = 16907 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 4364 Kgf	or W3 = 16907 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.8202 mm.

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Input, Nozzle Desc: K2A-2"

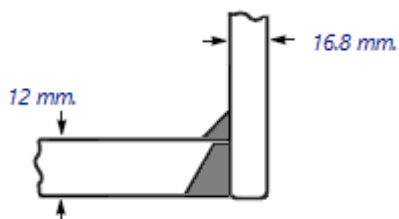
From: 20

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	1646.3336	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		450.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
= 6.6373 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.57}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0895 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

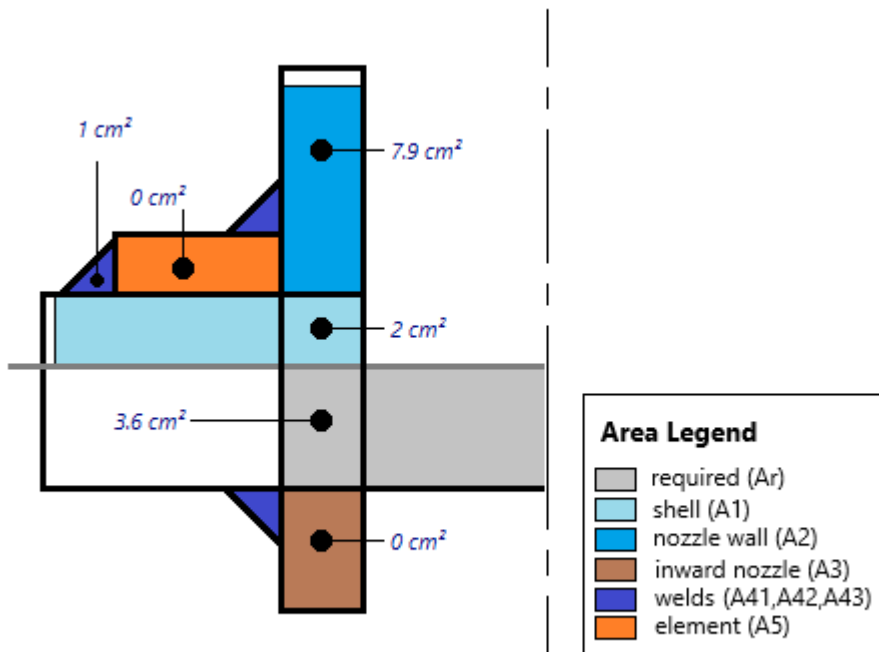


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.584	1.832	NA
Area in Shell	A1	2.032	1.952	NA
Area in Nozzle Wall	A2	7.857	7.683	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.889	10.635	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 6.6373 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.6373 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.584 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 1.0)$$

$$= 2.032 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.09)1.0$$

$$= 7.857 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.0258 mm.
Wall Thickness per UG16(b),	tr16b = 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.2373 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.2373 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3503 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm.
Wall Thickness per table UG-45	tb3 = 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.4, \max(8.2373, 4.1)]$$

$$= 6.4000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.0258, 6.4)$$

$$= 6.4000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 12.0, tr = 6.637, c = 1.6 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.638, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.57/51.00 = 0.090$$

Weld Size Calculations, Description: K2A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 10.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (3.5841 - 2.0319 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$$

$$= 3793.88 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (7.8575 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 12463.84 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$



$$= (7.8575 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (7.8575 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 3793 Kgf	or W1 = 12463 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 3793 Kgf	or W2 = 16912 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 3793 Kgf	or W3 = 16912 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0932 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 54.00 mm.

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Input, Nozzle Desc: N3-4"

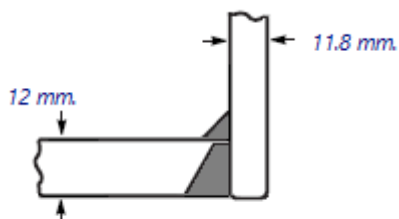
From: 20

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	1646.3336	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1050.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
 $= 6.6373 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.57 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 4.57)$
 $= 0.2208 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm.

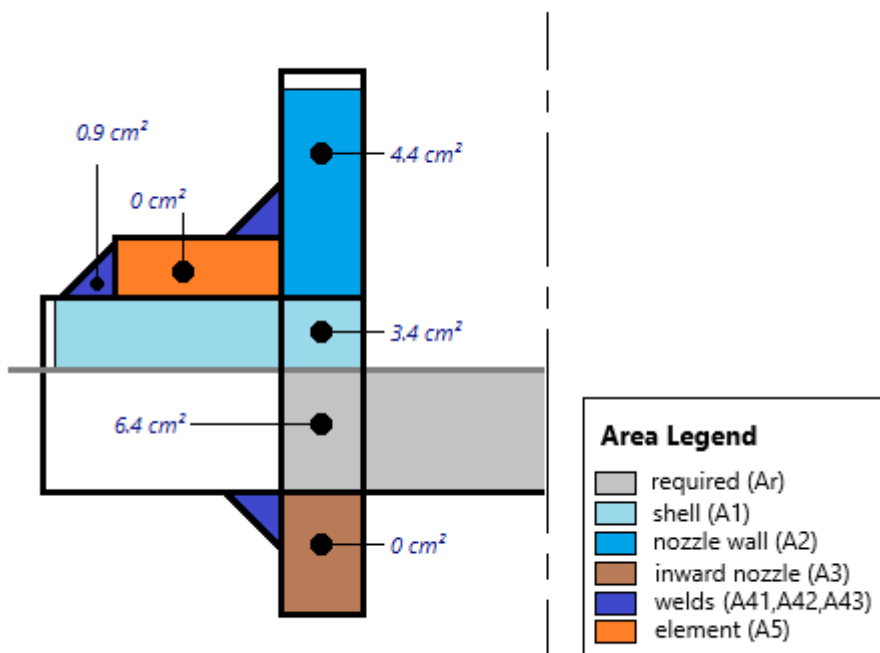
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.7941 mm.
 Parallel to Vessel Wall, opening length d 93.8970 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 25.5037 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.428	3.286	NA
Area in Shell	A1	3.422	3.288	NA
Area in Nozzle Wall	A2	4.353	4.242	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.630	8.385	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (93.897 \cdot 6.6373 \cdot 1.0 + 2 \cdot 10.2015 \cdot 6.6373 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.428 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.897(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 10.201$
 $(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 0.855)$
 $= 3.422 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 25.5)(10.2 - 0.22)0.8551$
 $= 4.353 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm.}$
Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure $trb1 = 8.2373 \text{ mm.}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 8.2373 \text{ mm.}$
Wall Thickness, shell/head, external pressure $trb2 = 2.3503 \text{ mm.}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.2373, 4.1)]$
 $= 6.8578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm.}$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.0,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 267.0 N./mm ²	Passed
Occasional	: 0.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 18.9,	Allowable	: 82.6 N./mm ²	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.57/51.00 = 0.090$

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.2015 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.4284 - 3.4218 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$
 $= 5154.52 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 7328.78 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.3531 + 0.0 + 0.8551 + 0.0 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 10586. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 52.0493) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118$$

$$= 14050. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 19443. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10586 + 14050) = 24636 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10586 + 0 + 19443 + 0) = 30029 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (10586 + 19443 + 0) = 30029 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 24635 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W1 = 7328 \text{ Kgf}$$

$$\text{Path 2-2} = 30029 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W2 = 9881 \text{ Kgf}$$

$$\text{Path 3-3} = 30029 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W3 = 9881 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8166 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.9647 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 93.90 mm.

Input Echo, WRC297 Item 1, Description: N3-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4000.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	114.300	mm.
Nozzle Thickness	11.8015	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²



Nozzle Hot S.I. Allowable SNmh 118.00 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.57	bars
Radial Load	P	764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (9.685) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2006.8(12.0 - 1.6))}$$

$$= 591.041 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4013.600 mm.
Nozzle Outside Diameter	(d) =	114.300 mm.
Vessel Thickness used	(T) =	10.400 mm.
Nozzle Thickness used	(t) =	10.201 mm.
T / t	=	1.019
d / t	=	11.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.559$$

Nr/P	=	0.163
Mr/P	=	0.262
M0/P	=	0.093
N0/P	=	0.171
MrD/Mc	=	0.288
NrDL/Mc	=	0.141
M0d/Mc	=	0.117
N0DL/Mc	=	-0.091
MrD/Ml	=	0.252
NrDL/Ml	=	0.134
M0D/Ml	=	0.100
N0DL/Ml	=	0.015

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-67	67	67	-67
Normal Pressure Stress	87	88	87	88
Outplane Stress Summary	-32	181	106	48

**Vessel Stresses (N./mm²):**

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-171	171	171	-171
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	-262	297	109	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9
Shear Stress Summary	12	12	6	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	263	298	115	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	10	10	-10	-10
Outplane Bending (Mc)	-79	79	79	-79
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	-75	160	62	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-195	195	195	-195
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	87	88	87	88
Inplane Stress Summary	-244	365	179	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	244	365	180	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-11	-10	-7	-7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-130	130	130	-130
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-227	190	68	-36
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	12	12	6	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	228	202	77	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	10	10	-10	-10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	0	1	-19	-19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-153	153	153	-153
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-250	213	91	-59

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	251	213	114	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	88	87	88	87	88	87	88
Circ. Pl (SUS)	-12	-12	-10	-10	-26	-26	4	4
Circ. Q (SUS)	-105	105	29	-29	-304	304	86	-86
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-26	-26	4	4	-1	-1	-21	-21
Long. Q (SUS)	-280	280	62	-62	-117	117	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	87.0	88.0	87.0	88.0	87.0	88.0	87.0	88.0
Pm+Pl (SUS)	75.1	76.1	77.1	78.1	61.2	62.2	91.1	92.1
Pm+Pl+Q (Total)	263.5	298.0	114.7	65.5	243.3	366.2	178.1	33.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	92.06	207.00	Passed

Pm+Pl+Q (TOTAL)	366.24	413.99	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-12	-12	-10	-10	-1	-1	-21	-21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-19	-19	15	15	-19	-19	15	15
Long. Q (SUS)	-208	208	52	-52	-231	231	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	18.5	18.5	24.3	24.3	19.4	19.4	35.2	35.2
Pm+Pl+Q (Total)	226.6	201.2	77.3	37.7	250.4	213.2	112.2	61.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	35.23	177.00	Passed
Pm+Pl+Q (TOTAL)	250.39	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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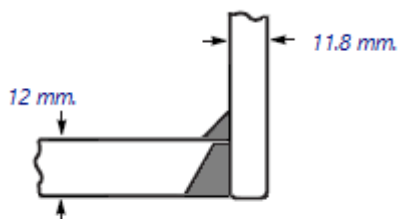
**Input, Nozzle Desc: N4-4"****From: 20**

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2624.9995	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	468.2500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4-4"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
 $= 6.6373 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.57 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 4.57)$
 $= 0.2208 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4996 mm.

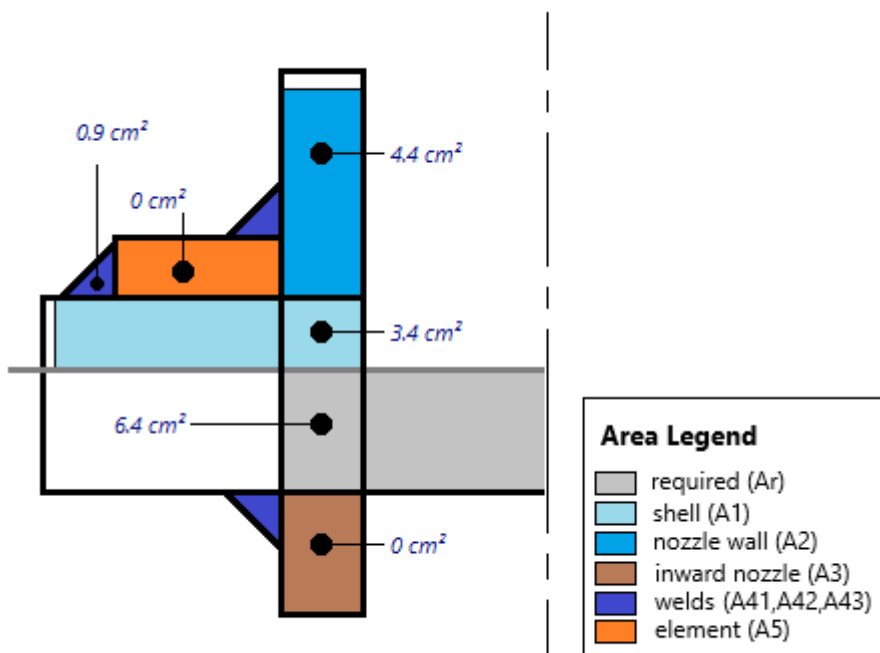
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.7941 mm.
 Parallel to Vessel Wall, opening length d 93.8970 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 25.5037 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.428	3.984	NA
Area in Shell	A1	3.422	1.976	NA
Area in Nozzle Wall	A2	4.353	4.232	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.630	7.062	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (93.897 \cdot 6.6373 \cdot 1.0 + 2 \cdot 10.2015 \cdot 6.6373 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.428 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.897(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 10.201$
 $(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 0.855)$
 $= 3.422 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 25.5)(10.2 - 0.22)0.8551$
 $= 4.353 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0996 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.2373 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.2373 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3503 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.2373, 4.1)]$
 $= 6.8578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0996, 6.8578)$
 $= 6.8578 \text{ mm.}$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.0,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 267.0 N./mm ²	Passed
Occasional	: 0.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 18.9,	Allowable	: 82.6 N./mm ²	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.57/51.00 = 0.090$

Weld Size Calculations, Description: N4-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.2015 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.4284 - 3.4218 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$
 $= 5154.52 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 7328.78 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.3531 + 0.0 + 0.8551 + 0.0 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 10586. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 52.0493) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118$
 $= 14050. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 19443. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (10586 + 14050) = 24636 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (10586 + 0 + 19443 + 0) = 30029 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (10586 + 19443 + 0) = 30029 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 24635	Kgf, must exceed W = 5154 Kgf	or W1 = 7328 Kgf
Path 2-2 = 30029	Kgf, must exceed W = 5154 Kgf	or W2 = 9881 Kgf
Path 3-3 = 30029	Kgf, must exceed W = 5154 Kgf	or W3 = 9881 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8166 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 481.0667 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 93.90 mm.

Input Echo, WRC297 Item 1, Description: N4-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4000.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	114.300	mm.
Nozzle Thickness	11.8015	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²



Nozzle Hot S.I. Allowable SNmh 118.00 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4.57	bars
Radial Load	P	764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (9.685) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(2006.8(12.0 - 1.6))}$$

$$= 591.041 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4013.600 mm.
Nozzle Outside Diameter	(d) =	114.300 mm.
Vessel Thickness used	(T) =	10.400 mm.
Nozzle Thickness used	(t) =	10.201 mm.
T / t	=	1.019
d / t	=	11.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.559$$

Nr/P	=	0.163
Mr/P	=	0.262
M0/P	=	0.093
N0/P	=	0.171
MrD/Mc	=	0.288
NrDL/Mc	=	0.141
M0d/Mc	=	0.117
N0DL/Mc	=	-0.091
MrD/Ml	=	0.252
NrDL/Ml	=	0.134
M0D/Ml	=	0.100
N0DL/Ml	=	0.015

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-67	67	67	-67
Normal Pressure Stress	87	88	87	88
Outplane Stress Summary	-32	181	106	48

**Vessel Stresses (N./mm²):**

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-171	171	171	-171
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	-262	297	109	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9
Shear Stress Summary	12	12	6	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	263	298	115	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	10	10	-10	-10
Outplane Bending (Mc)	-79	79	79	-79
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	-75	160	62	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-195	195	195	-195
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	87	88	87	88
Inplane Stress Summary	-244	365	179	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	244	365	180	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-11	-10	-7	-7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-130	130	130	-130
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-227	190	68	-36
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	12	12	6	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	228	202	77	37
------------------------	-----	-----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	10	10	-10	-10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	0	1	-19	-19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-153	153	153	-153
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-250	213	91	-59

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	251	213	114	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	88	87	88	87	88	87	88
Circ. Pl (SUS)	-12	-12	-10	-10	-26	-26	4	4
Circ. Q (SUS)	-105	105	29	-29	-304	304	86	-86
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-26	-26	4	4	-1	-1	-21	-21
Long. Q (SUS)	-280	280	62	-62	-117	117	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	87.0	88.0	87.0	88.0	87.0	88.0	87.0	88.0
Pm+Pl (SUS)	75.1	76.1	77.1	78.1	61.2	62.2	91.1	92.1
Pm+Pl+Q (Total)	263.5	298.0	114.7	65.5	243.3	366.2	178.1	33.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	92.06	207.00	Passed



Pm+Pl+Q (TOTAL)	366.24	413.99	Passed
-----------------	--------	--------	--------

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-12	-12	-10	-10	-1	-1	-21	-21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-19	-19	15	15	-19	-19	15	15
Long. Q (SUS)	-208	208	52	-52	-231	231	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	18.5	18.5	24.3	24.3	19.4	19.4	35.2	35.2
Pm+Pl+Q (Total)	226.6	201.2	77.3	37.7	250.4	213.2	112.2	61.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	35.23	177.00	Passed
Pm+Pl+Q (TOTAL)	250.39	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K3-2"

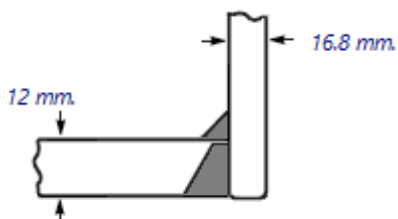
From: 30

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2624.9995	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3450.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	430.1500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
= 6.6373 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.57}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0895 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4062 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

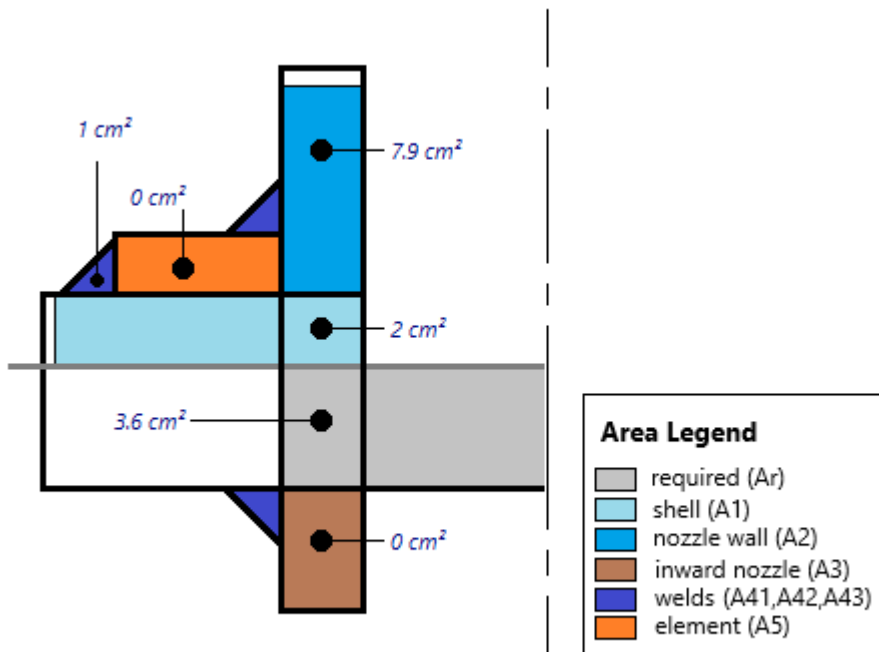


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.584	2.221	NA
Area in Shell	A1	2.032	1.173	NA
Area in Nozzle Wall	A2	7.857	7.693	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.889	9.866	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 6.6373 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.6373 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.584 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 1.0)$$

$$= 2.032 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.09)1.0$$

$$= 7.857 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 1.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0062$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.2373$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.2373$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 2.3503$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb_3 , max(tb_1 , tb_2)]
 = min[6.4, max(8.2373, 4.1)]
 = 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(t_a , t_b)
 = max(2.0062, 6.4)
 = 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 16.8$, $tr = 0.09$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.006$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $tr = 6.637$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.638$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $4.57/51.00 = 0.090$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.4000$ mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2* t_n * f_r *(E_1 * t - tr)) S_v)
 = max(0, (3.5841 - 2.0319 + 2 * 15.2 * 1.0 *
 (1.0 * 10.4 - 6.6373))138)
 = 3793.88 Kg

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (7.8575 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 12463.84 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (7.8575 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (7.8575 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 3793 Kgf	or W1 = 12463 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 3793 Kgf	or W2 = 16912 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 3793 Kgf	or W3 = 16912 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 442.5952 mm.

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Input, Nozzle Desc: K4-2"

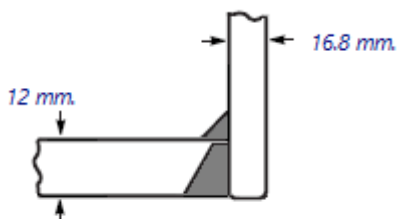
From: 30

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2622.0000	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4050.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
= 6.6373 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.57}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0895 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

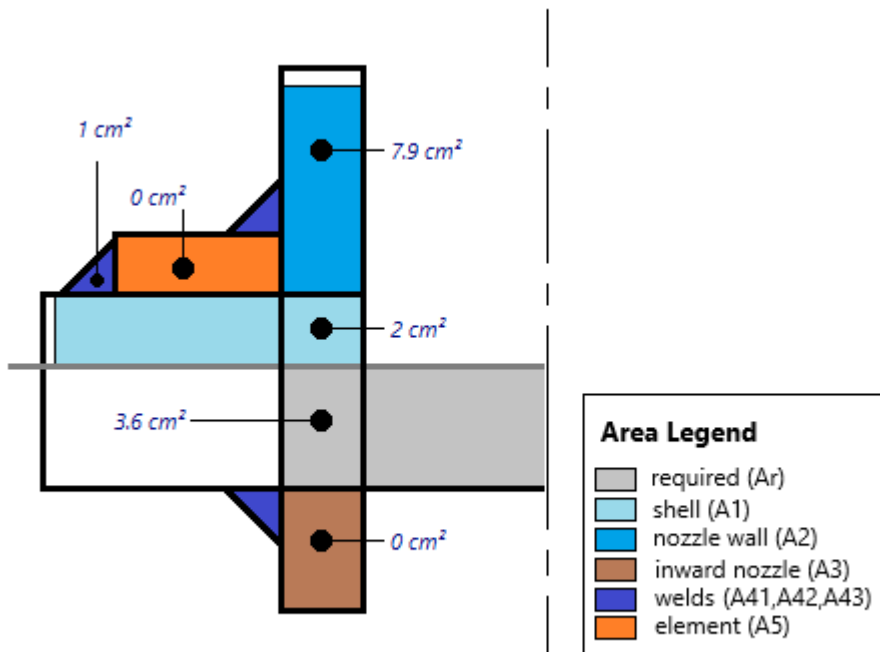


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.584	2.220	NA
Area in Shell	A1	2.032	1.175	NA
Area in Nozzle Wall	A2	7.857	7.683	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.889	9.858	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 6.6373 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.6373 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.584 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0 (1.0 \cdot 10.4 - 1.0 \cdot 6.6373) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 1.0)$$

$$= 2.032 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 26.0) (15.2 - 0.09) 1.0$$

$$= 7.857 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.0258 mm.
Wall Thickness per UG16(b),	tr16b = 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.2373 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.2373 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3503 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm.
Wall Thickness per table UG-45	tb3 = 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.4, \max(8.2373, 4.1)]$$

$$= 6.4000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.0258, 6.4)$$

$$= 6.4000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 12.0, tr = 6.637, c = 1.6 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.638, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.57/51.00 = 0.090$$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 10.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (3.5841 - 2.0319 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$$

$$= 3793.88 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (7.8575 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 12463.84 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$



$$= (7.8575 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (7.8575 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 3793 Kgf	or W1 = 12463 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 3793 Kgf	or W2 = 16912 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 3793 Kgf	or W3 = 16912 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0932 mm.

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Input, Nozzle Desc: N1-18

From: 40

Pressure for Reinforcement Calculations	P	4.925	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2725.0002	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.

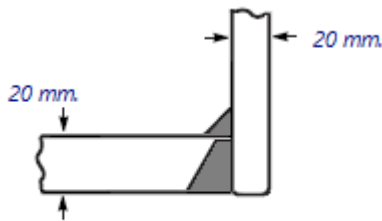
Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1000.0000	mm.
Distance from Bottom/Left Tangent		7050.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	750.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-18**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.93 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.93)$
 $= 7.1597 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.93 \cdot 228.6 / (138 \cdot 1.0 + 0.4 \cdot 4.93)$
 $= 0.8148 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3587 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	840.8000 mm.
Parallel to Vessel Wall, opening length	d	420.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	46.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

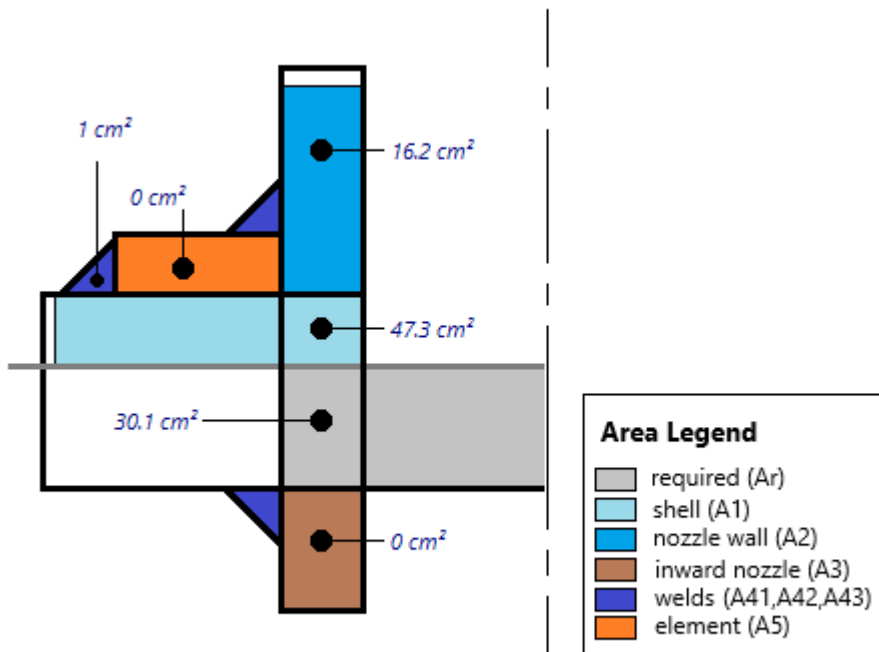


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	30.099	17.545	NA
Area in Shell	A1	47.254	42.264	NA
Area in Nozzle Wall	A2	16.178	15.678	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	64.433	58.942	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (420.4 \cdot 7.1597 \cdot 1.0 + 2 \cdot 18.4 \cdot 7.1597 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 30.099 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 420.4 (1.0 \cdot 18.4 - 1.0 \cdot 7.16) - 2 \cdot 18.4$$

$$(1.0 \cdot 18.4 - 1.0 \cdot 7.1597) \cdot (1 - 1.0)$$

$$= 47.254 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 46.0) (18.4 - 0.81) 1.0$$

$$= 16.178 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 1.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.9587$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.7597$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.7597$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 2.3503$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 9.9312$ mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb_3 , max(tb_1 , tb_2)]
 = min[9.931, max(8.7597, 4.1)]
 = 8.7597 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(t_a , t_b)
 = max(2.9587, 8.7597)
 = 8.7597 mm.

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	19.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	326.0 N./mm ²	Passed
Occasional	:	2.7,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	10.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $tr = 0.815$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.044$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 20.0$, $tr = 7.16$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.389$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $4.93/19.60 = 0.251$

Weld Size Calculations, Description: N1-18

Intermediate Calc. for nozzle/shell Welds	T_{min}	18.4000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (30.0994 - 47.2542 + 2 \cdot 18.4 \cdot 1.0 \cdot (1.0 \cdot 18.4 - 7.1597)) \cdot 138) \\ &= \max(0, -18318.78) \text{ Kg} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (16.1784 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138 \\ &= 24172.71 \text{ Kg} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (16.1784 + 0.0 + 1. + (6.7712)) \cdot 138 \\ &= 33700.86 \text{ Kg} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (16.1784 + 0.0 + 1. + 0.0 + (6.7712)) \cdot 138 \\ &= 33700.86 \text{ Kg} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 457.2 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 49518. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 219.4) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138 \\ &= 124926. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 457.2 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 89741. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (49518 + 124926) = 174444 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (49518 + 0 + 89741 + 0) = 139259 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (49518 + 89741 + 0) = 139259 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 174444 Kg, must exceed W = 0 Kg	or W1 = 24172 Kg
Path 2-2 = 139258 Kg, must exceed W = 0 Kg	or W2 = 33700 Kg
Path 3-3 = 139258 Kg, must exceed W = 0 Kg	or W3 = 33700 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 13.1073 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 783.1073 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.575 %

Input Echo, WRC297 Item 1, Description: N1-18 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.6000	mm.	
Shell External Corrosion Allowance	0.0000	mm.	
Shell Diameter	4000.000	mm.	
Shell Thickness	20.0000	mm.	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	1.6000	mm.	
Nozzle Diameter	457.200	mm.	
Nozzle Thickness	20.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	4.93	bars
Radial Load	P	3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(2010.8(20.0 - 1.6))}$$

$$= 1091.957 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4021.600 mm.
Nozzle Outside Diameter	(d) =	457.200 mm.
Vessel Thickness used	(T) =	18.400 mm.
Nozzle Thickness used	(t) =	18.400 mm.
T / t	=	1.000
d / t	=	24.848

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.681$$



Nr/P = 0.061
 Mr/P = 0.141
 M0/P = 0.049
 N0/P = 0.124
 MrD/Mc = 0.262
 NrDL/Mc = 0.095
 M0d/Mc = 0.096
 N0DL/Mc = 0.102
 MrD/Ml = 0.140
 NrDL/Ml = 0.064
 M0D/Ml = 0.048
 N0DL/Ml = 0.181

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-29	29	-29	29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-33	-33	33	33
Outplane Bending (ML)	-52	52	52	-52
Normal Pressure Stress	53	53	53	53
Outplane Stress Summary	-73	89	97	51

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-84	84	-84	84
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-154	154	154	-154
Inplane Pressure Stress	26	26	26	26
Inplane Stress Summary	-229	247	101	-37

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	230	248	105	88

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-29	29	-29	29



Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-105	105	105	-105
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	-138	130	109	-43
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-84	84	-84	84
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-287	287	287	-287
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	53	53	53

Inplane Stress Summary	-342	402	267	-138
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	342	402	268	138
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-12	-12	-12	-12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-33	-33	33	33
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	-38	-38	27	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-66	66	-66	66
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-118	118	118	-118
Axial Pressure Stress	3	3	3	3



Axial Stress Summary -193 176 64 -39

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary 8 8 4 4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	194	215	65	68

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-12	-12	-12	-12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary -24 -23 12 13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-66	66	-66	66
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-235	235	235	-235
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary -310 293 181 -156

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary 4 4 8 8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	310	317	182	170

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of

Stress Values at



Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	53	53	53	53	53	53	53
Circ. Pl (SUS)	-45	-45	21	21	-23	-23	11	11
Circ. Q (SUS)	-81	81	23	-23	-371	371	203	-203
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-17	-17	5	5	-30	-30	6	6
Long. Q (SUS)	-238	238	70	-70	-134	134	76	-76
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	53.0	53.0	53.0	53.0	53.0	53.0	53.0	53.0
Pm+Pl (SUS)	9.6	9.6	74.0	74.0	34.1	34.1	64.0	64.0
Pm+Pl+Q (Total)	229.3	247.3	104.4	90.6	341.1	401.1	267.3	139.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	74.02	207.00	Passed
Pm+Pl+Q (TOTAL)	401.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-45	-45	21	21	-30	-30	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-11	-11	9	9	-11	-11	9	9
Long. Q (SUS)	-184	184	52	-52	-301	301	169	-169
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	39.0	38.0	27.1	28.1	24.1	23.1	13.0	13.6
Pm+Pl+Q (Total)	192.3	214.5	64.7	68.7	309.1	316.2	181.3	170.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	39.03	207.00	Passed
Pm+Pl+Q (TOTAL)	316.16	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

CCW Expansion Vessel (V8804)

PV Elite 28

Licensee: SL User



FileName : CCW Expansion Vessel (V8804)-7%

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Nozzle Calcs.: N1-18

Nozl: 36 3:03pm Sep 16,2026



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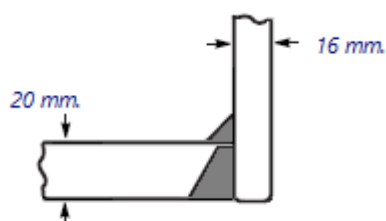
**Input, Nozzle Desc: N2-8"****Parent: N1-18**

Pressure for Reinforcement Calculations	P	4.925	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	417.20	mm.
Design Length of Section	L	750.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 8.625 in.
Actual Thickness Used in Calculation 0.629 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.93 \cdot 210.2 / (138 \cdot 1.0 - 0.6 \cdot 4.93)$
= 0.7519 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $4.93 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 4.93)$
= 0.4565 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5132 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	380.6309	mm.
Parallel to Vessel Wall, opening length	d	190.3154	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	35.9494	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 118.0/138.0)
= 0.855

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 118.0/138.0)
= 0.855

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.855, 1.0)
= 0.855

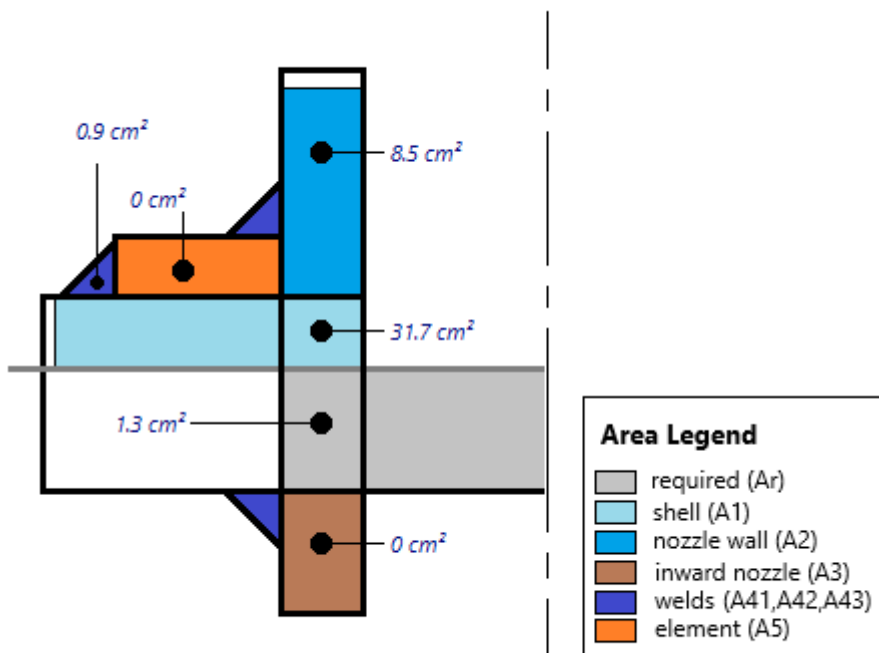


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	1.462	1.321	NA
Area in Shell	A1	32.852	31.722	NA
Area in Nozzle Wall	A2	8.560	8.525	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	42.266	41.102	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(190.3154 \cdot 1.3587 \cdot 1 + 2 \cdot 14.3798 \cdot 1.3587 \cdot 1(1-0.86))$
 $= 1.321 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 190.315(1.0 \cdot 18.4 - 1.0 \cdot 1.359) - 2 \cdot 14.38$
 $(1.0 \cdot 18.4 - 1.0 \cdot 1.3587) \cdot (1 - 0.855)$
 $= 31.722 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 35.95)(14.38 - 0.51) 0.8551$
 $= 8.525 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.1132 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 2.3519 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 4.1000 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 8.7600 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.76, \max(4.1, 4.1)]$
 $= 4.1000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.1132, 4.1)$
 $= 4.1000 \text{ mm.}$

Available Nozzle Neck Thickness = 15.9798 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	274.0 N./mm ²	Passed
Occasional	:	1.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	13.1,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, tg = 15.98, tr = 0.456, c = 1.6 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 15.98, tr = 0.456, c = 1.6 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.032$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $4.93/19.60 = 0.251$ Weld Size Calculations, Description: N2-8"

Intermediate Calc. for nozzle/shell Welds Tmin 14.3798 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (1.3212 - 31.7219 + 2 \cdot 14.3798 \cdot 0.8551 \cdot (1.0 \cdot 18.4 - 1.3587)) \cdot 138)$$

$$= \max(0, -36882.19) \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (8.525 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$$

$$= 13199.50 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (8.525 + 0.0 + 0.8551 + (4.5248)) \cdot 138$$

$$= 19566.77 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (8.525 + 0.0 + 0.8551 + 0.0 + (4.5248)) \cdot 138$$

$$= 19566.77 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 20289. \text{ Kg}$$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 102.3476) \cdot (15.9798 - 1.6) \cdot 0.7 \cdot 118$$

$$= 38943. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gwnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 43001. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (20289 + 38943) = 59232 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20289 + 0 + 43001 + 0) = 63290 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (20289 + 43001 + 0) = 63290 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 59232	Kgf, must exceed W = 0 Kg	or W1 = 13199 Kg
Path 2-2 = 63289	Kgf, must exceed W = 0 Kg	or W2 = 19566 Kg
Path 3-3 = 63289	Kgf, must exceed W = 0 Kg	or W3 = 19566 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 67 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 31.0740 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.0740 mm.

Input Echo, WRC297 Item 1, Description: N2-8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	417.200	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	219.075	mm.
Nozzle Thickness	15.9798	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	4.93	bars
Radial Load	P	1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{219.4(20.0 - 1.6)}$$

$$= 428.747 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	438.800 mm.
Nozzle Outside Diameter	(d) =	219.075 mm.
Vessel Thickness used	(T) =	18.400 mm.
Nozzle Thickness used	(t) =	14.380 mm.
T / t	=	1.280
d / t	=	15.235

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.438$$

Nr/P	=	0.078
Mr/P	=	0.109
M0/P	=	0.042
N0/P	=	0.120
MrD/Mc	=	0.234
NrDL/Mc	=	0.131
M0d/Mc	=	0.103
N0DL/Mc	=	0.087
MrD/Ml	=	0.120
NrDL/Ml	=	0.074
M0D/Ml	=	0.056
N0DL/Ml	=	0.161

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-12	-12	12	12
Outplane Bending (Ml)	-25	25	25	-25
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-48	24	26	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-28	28	-28	28
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	2	2	2	2

Inplane Stress Summary	-89	76	30	-20
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	90	77	33	21
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2

Outplane Stress Summary	-67	48	39	-31
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-28	28	-28	28
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-105	105	105	-105
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	5	5	5	5

Inplane Stress Summary	-142	127	88	-64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress	142	127	89	66
------------------------	-----	-----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	-13	-13	10	11
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-30	30	-30	30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-61	61	61	-61
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-104	79	43	-18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	105	94	44	31
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	-8	-7	4	5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-30	30	-30	30
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-124	124	124	-124
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-167	142	107	-81
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	167	150	108	89

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	5	5	5	5	5	5	5
Circ. Pl (SUS)	-17	-17	7	7	-12	-12	6	6
Circ. Q (SUS)	-36	36	14	-14	-133	133	77	-77
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-8	-8	2	2	-11	-11	1	1
Long. Q (SUS)	-82	82	26	-26	-57	57	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Pm+Pl (SUS)	12.2	12.2	12.1	12.1	9.4	9.4	11.1	11.1
Pm+Pl+Q (Total)	88.9	76.7	32.5	22.8	140.2	126.2	88.7	67.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	12.16	207.00	Passed
Pm+Pl+Q (TOTAL)	140.22	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	-17	-17	7	7	-11	-11	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-13	-13	11	11	-13	-13	11	11
Long. Q (SUS)	-91	91	31	-31	-154	154	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	15.2	14.6	13.2	13.6	12.8	12.7	12.5	12.5
Pm+Pl+Q (Total)	103.9	93.7	43.7	31.6	166.2	149.3	106.8	88.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	15.24	177.00	Passed
Pm+Pl+Q (TOTAL)	166.16	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-4"

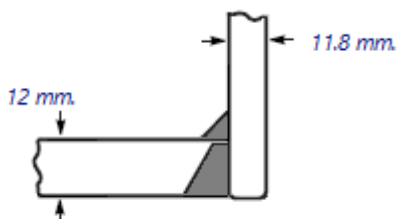
From: 40

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2725.0002	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7250.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N5-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
 $= 6.6373 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.57 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 4.57)$
 $= 0.2208 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	187.7941 mm.
Parallel to Vessel Wall, opening length	d	93.8970 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.5037 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

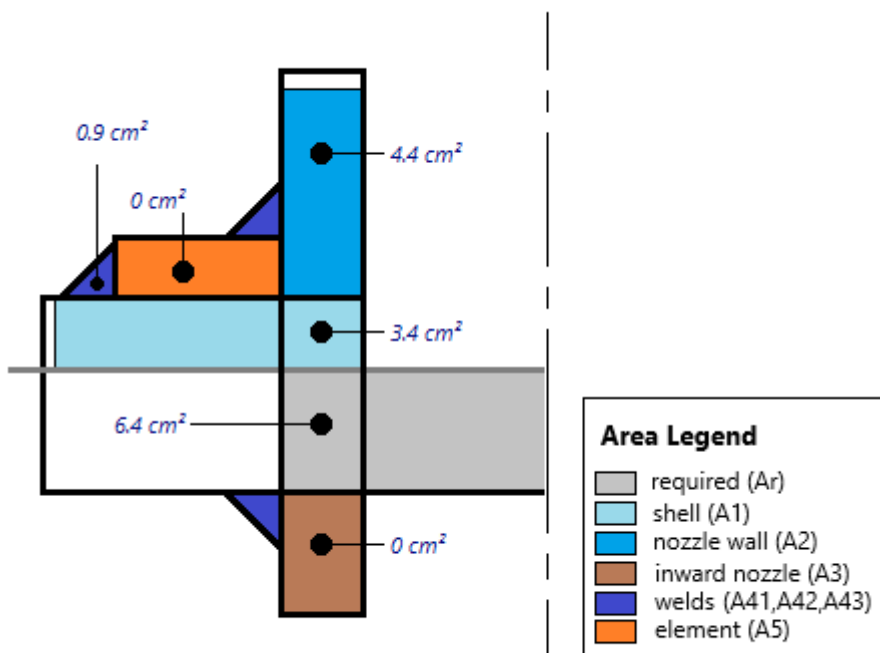


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.428	4.042	NA
Area in Shell	A1	3.422	1.867	NA
Area in Nozzle Wall	A2	4.353	4.242	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.630	6.965	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (93.897 \cdot 6.6373 \cdot 1.0 + 2 \cdot 10.2015 \cdot 6.6373 \cdot 1.0 \cdot (1 - 0.86))$
 $= 6.428 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.897(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 10.201$
 $(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 0.855)$
 $= 3.422 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 25.5)(10.2 - 0.22)0.8551$
 $= 4.353 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm.}$
Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure $trb1 = 8.2373 \text{ mm.}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 8.2373 \text{ mm.}$
Wall Thickness, shell/head, external pressure $trb2 = 2.3503 \text{ mm.}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.858, \max(8.2373, 4.1)]$
 $= 6.8578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.0751, 6.8578)$
 $= 6.8578 \text{ mm.}$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.0 N./mm ²	Passed
Occasional	:	0.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	18.9,	Allowable	:	82.6 N./mm ²	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.57/51.00 = 0.090$

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.2015 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.4284 - 3.4218 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$
 $= 5154.52 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 7328.78 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.3531 + 0.0 + 0.8551 + 0.0 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 10586. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 52.0493) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118$$

$$= 14050. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 19443. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10586 + 14050) = 24636 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10586 + 0 + 19443 + 0) = 30029 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (10586 + 19443 + 0) = 30029 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 24635 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W1 = 7328 \text{ Kgf}$$

$$\text{Path 2-2} = 30029 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W2 = 9881 \text{ Kgf}$$

$$\text{Path 3-3} = 30029 \text{ Kgf, must exceed } W = 5154 \text{ Kgf} \quad \text{or } W3 = 9881 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8166 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.9647 mm.

Input Echo, WRC297 Item 1, Description: N5-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4000.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	114.300	mm.
Nozzle Thickness	11.8015	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	4.57	bars
Radial Load	P	764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.685) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2006.8(12.0 - 1.6)}$$

$$= 591.041 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4013.600 mm.
Nozzle Outside Diameter	(d) =	114.300 mm.
Vessel Thickness used	(T) =	10.400 mm.
Nozzle Thickness used	(t) =	10.201 mm.
T / t	=	1.019
d / t	=	11.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.559$$

Nr/P	=	0.163
Mr/P	=	0.262
M0/P	=	0.093
N0/P	=	0.171
MrD/Mc	=	0.288
NrDL/Mc	=	0.141
M0d/Mc	=	0.117
N0DL/Mc	=	-0.091
MrD/Ml	=	0.252
NrDL/Ml	=	0.134
M0D/Ml	=	0.100
N0DL/Ml	=	0.015

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-67	67	67	-67
Normal Pressure Stress	87	88	87	88
Outplane Stress Summary	-32	181	106	48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-171	171	171	-171
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	-262	297	109	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9
Shear Stress Summary	12	12	6	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	263	298	115	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-11	-11	-11	-11
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	10	10	-10	-10
Outplane Bending (Mc)	-79	79	79	-79
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	-75	160	62	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-109	109	-109	109
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-195	195	195	-195
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	87	88	87	88
Inplane Stress Summary	-244	365	179	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9



Shear Stress Summary	6	6	12	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	244	365	180	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-11	-10	-7	-7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-130	130	130	-130
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-227	190	68	-36
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	12	12	6	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	228	202	77	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-11	-11	-11	-11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	10	10	-10	-10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	0	1	-19	-19
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**Nozzle Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-78	78	-78	78
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-153	153	153	-153
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-250	213	91	-59

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	6	6	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	251	213	114	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	88	87	88	87	88	87	88
Circ. Pl (SUS)	-12	-12	-10	-10	-26	-26	4	4
Circ. Q (SUS)	-105	105	29	-29	-304	304	86	-86
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-26	-26	4	4	-1	-1	-21	-21
Long. Q (SUS)	-280	280	62	-62	-117	117	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	87.0	88.0	87.0	88.0	87.0	88.0	87.0	88.0
Pm+Pl (SUS)	75.1	76.1	77.1	78.1	61.2	62.2	91.1	92.1
Pm+Pl+Q (Total)	263.5	298.0	114.7	65.5	243.3	366.2	178.1	33.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	92.06	207.00	Passed
Pm+Pl+Q (TOTAL)	366.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-12	-12	-10	-10	-1	-1	-21	-21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-19	-19	15	15	-19	-19	15	15
Long. Q (SUS)	-208	208	52	-52	-231	231	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	18.5	18.5	24.3	24.3	19.4	19.4	35.2	35.2
Pm+Pl+Q (Total)	226.6	201.2	77.3	37.7	250.4	213.2	112.2	61.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	35.23	177.00	Passed
Pm+Pl+Q (TOTAL)	250.39	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N6-4"

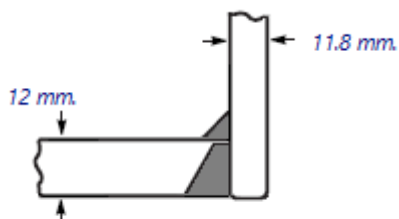
From: 40

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	2725.0002	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7650.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	414.1480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N6-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
 $= 6.6373 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.57 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 4.57)$
 $= 0.2208 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4751 mm.

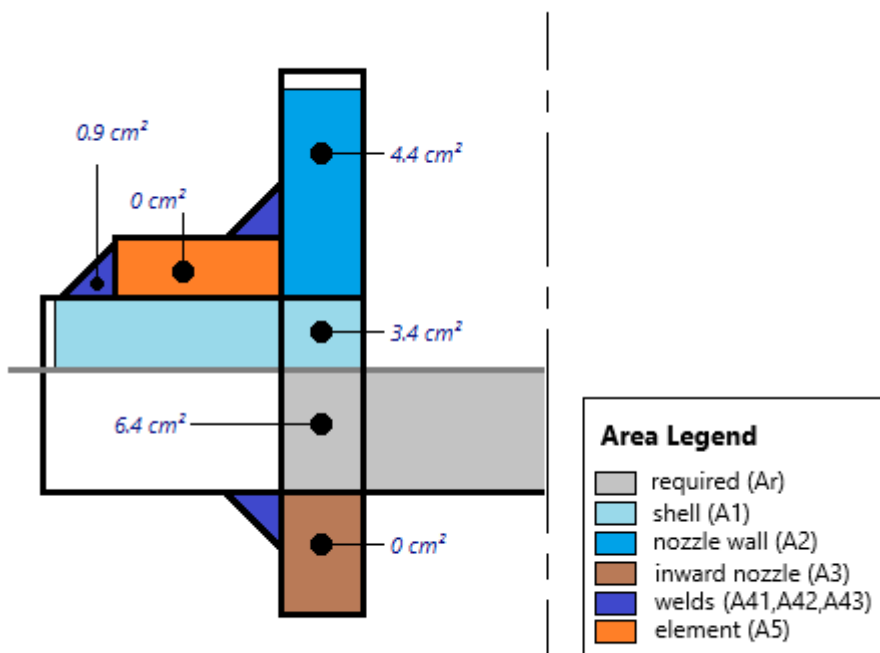
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.7941 mm.
 Parallel to Vessel Wall, opening length d 93.8970 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 25.5037 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

**Figure UG 37.1 Nozzle Areas**

Results of Nozzle Reinforcement Area Calculations: (cm²)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6.428	4.042	NA
Area in Shell	A1	3.422	1.867	NA
Area in Nozzle Wall	A2	4.353	4.242	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	8.630	6.965	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (93.897 \cdot 6.6373 \cdot 1.0 + 2 \cdot 10.2015 \cdot 6.6373 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 6.428 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 93.897(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 10.201$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 0.855)$$

$$= 3.422 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 25.5)(10.2 - 0.22)0.8551$$

$$= 4.353 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.0751 \text{ mm.}$

Wall Thickness per UG16(b), $tr16b = 4.1000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb1 = 8.2373 \text{ mm.}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 8.2373 \text{ mm.}$

Wall Thickness, shell/head, external pressure $trb2 = 2.3503 \text{ mm.}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.1000 \text{ mm.}$

Wall Thickness per table UG-45 $tb3 = 6.8578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.858, \max(8.2373, 4.1)]$$

$$= 6.8578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.0751, 6.8578)$$

$$= 6.8578 \text{ mm.}$$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	23.5,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	271.5 N./mm ²	Passed
Occasional	:	0.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	4.7,	Allowable	:	82.6 N./mm ²	Passed



The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 11.801$, $t_r = 0.221$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.57/51.00 = 0.090$

Weld Size Calculations, Description: N6-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.2015 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6.4284 - 3.4218 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$
 $= 5154.52 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 7328.78 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.3531 + 0.0 + 0.8551 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.3531 + 0.0 + 0.8551 + 0.0 + (1.8144)) \cdot 138$
 $= 9881.95 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 10586. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 52.0493) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118$$

$$= 14050. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 19443. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10586 + 14050) = 24636 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10586 + 0 + 19443 + 0) = 30029 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (10586 + 19443 + 0) = 30029 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 24635 \text{ Kg}, \text{ must exceed } W = 5154 \text{ Kg} \quad \text{or } W1 = 7328 \text{ Kg}$$

$$\text{Path 2-2} = 30029 \text{ Kg}, \text{ must exceed } W = 5154 \text{ Kg} \quad \text{or } W2 = 9881 \text{ Kg}$$

$$\text{Path 3-3} = 30029 \text{ Kg}, \text{ must exceed } W = 5154 \text{ Kg} \quad \text{or } W3 = 9881 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.8166 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.9647 mm.

Input Echo, WRC297 Item 1, Description: N6-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	4000.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	114.300	mm.
Nozzle Thickness	11.8015	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	4.57	bars
Radial Load	P	-764.84	Kgf
Circumferential Shear	Vc	-509.89	Kgf
Longitudinal Shear	Vl	-509.89	Kgf
Circumferential Moment	Mc	-142.71	Kg-m.
Longitudinal Moment	Ml	-142.71	Kg-m.
Torsional Moment	Mt	-203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.685) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{2006.8(12.0 - 1.6)}$$

$$= 591.041 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4013.600 mm.
Nozzle Outside Diameter	(d) =	114.300 mm.
Vessel Thickness used	(T) =	10.400 mm.
Nozzle Thickness used	(t) =	10.201 mm.
T / t	=	1.019
d / t	=	11.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.559$$

Nr/P	=	0.163
Mr/P	=	0.262
M0/P	=	0.093
N0/P	=	0.171
MrD/Mc	=	0.288
NrDL/Mc	=	0.141
M0d/Mc	=	0.117
N0DL/Mc	=	-0.091
MrD/Ml	=	0.252
NrDL/Ml	=	0.134
M0D/Ml	=	0.100
N0DL/Ml	=	0.015

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	1	1	-1	-1
Outplane Bending (Ml)	67	-67	-67	67
Normal Pressure Stress	87	88	87	88
Outplane Stress Summary	207	-4	68	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	109	-109	109	-109
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	15	15	-15	-15
Inplane Bending (ML)	171	-171	-171	171
Inplane Pressure Stress	43	43	43	43
Inplane Stress Summary	350	-210	-22	102

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	-2	-2	2	2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	-9	-9	-9	-9
Shear Stress Summary	-12	-12	-6	-6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	351	210	91	129

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	11	11	11	11
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	79	-79	-79	79
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	163	-72	24	107

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	109	-109	109	-109
Inplane Membrane (Mc)	15	15	-15	-15
Inplane Bending (Mc)	195	-195	-195	195
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	87	88	87	88
Inplane Stress Summary	419	-189	-3	170

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	2	2	-2	-2
Torsional Shear (Mt)	-9	-9	-9	-9



Shear Stress Summary	-6	-6	-12	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	420	189	37	172
------------------------	-----	-----	----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	15	16	12	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	78	-78	78	-78
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	17	17	-17	-17
Axial Bending (ML)	130	-130	-130	130
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	229	-187	-66	38
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	-2	-2	2	2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	-9	-9	-9	-9

Shear Stress Summary	-12	-12	-6	-6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	230	205	80	40
------------------------	-----	-----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	11	11	11	11
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	3	4	24	24
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**Nozzle Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	78	-78	78	-78
Axial Membrane (Mc)	17	17	-17	-17
Axial Bending (Mc)	153	-153	-153	153
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	252	-210	-89	61

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	2	2	-2	-2
Shear due to Torsion	-9	-9	-9	-9
Shear Stress Summary	-6	-6	-12	-12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	252	215	116	65

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	87	88	87	88	87	88	87	88
Circ. Pl (SUS)	12	12	10	10	26	26	-4	-4
Circ. Q (SUS)	105	-105	-29	29	304	-304	-86	86
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	26	26	-4	-4	1	1	21	21
Long. Q (SUS)	280	-280	-62	62	117	-117	-41	41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	-2	-2	2	2	2	2	-2	-2
Shear Q (SUS)	-9	-9	-9	-9	-9	-9	-9	-9
Pm (SUS)	87.0	88.0	87.0	88.0	87.0	88.0	87.0	88.0
Pm+Pl (SUS)	99.1	100.1	97.1	98.1	113.1	114.1	83.2	84.2
Pm+Pl+Q (Total)	349.8	211.6	92.1	128.8	417.2	190.4	34.1	171.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	114.06	207.00	Passed
Pm+Pl+Q (TOTAL)	417.19	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	12	12	10	10	1	1	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	19	19	-15	-15	19	19	-15	-15
Long. Q (SUS)	208	-208	-52	52	231	-231	-75	75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	-2	-2	2	2	2	2	-2	-2
Shear Q (SUS)	-9	-9	-9	-9	-9	-9	-9	-9
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	20.6	20.6	26.3	26.3	20.2	20.2	37.2	37.2
Pm+Pl+Q (Total)	228.6	203.2	79.2	39.8	251.2	214.5	114.1	64.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	37.22	177.00	Passed
Pm+Pl+Q (TOTAL)	251.20	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1A-2"

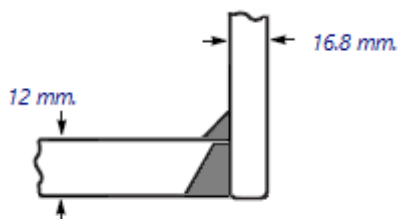
From: 50

Pressure for Reinforcement Calculations	P	4.567	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	1548.3328	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		9550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.57 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.57)$
= 6.6373 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.57}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0895 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4258 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

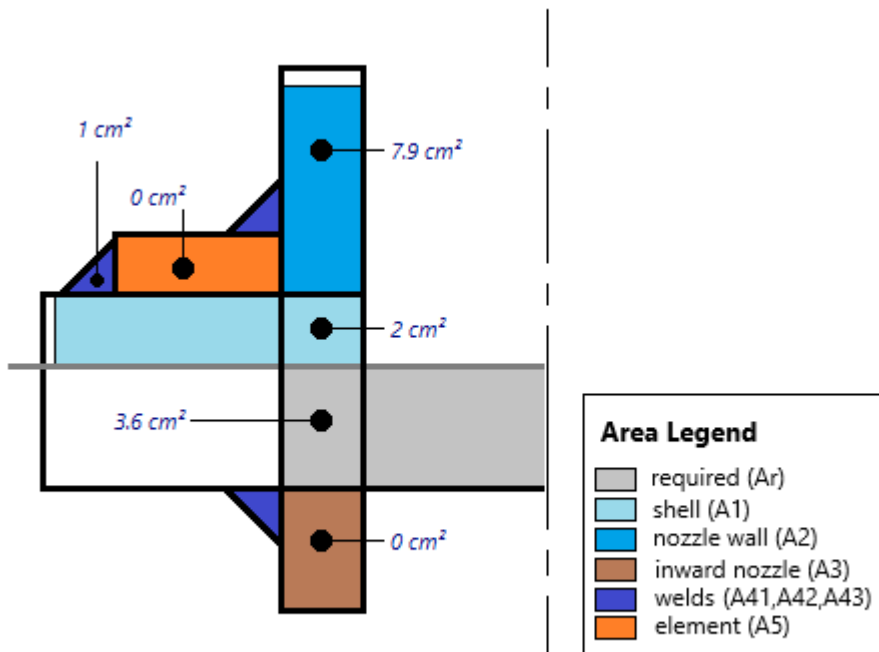


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.584	1.784	NA
Area in Shell	A1	2.032	2.047	NA
Area in Nozzle Wall	A2	7.857	7.683	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.889	10.730	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 6.6373 \cdot 1.0 + 2 \cdot 15.2 \cdot 6.6373 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.584 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 6.637) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 6.6373) \cdot (1 - 1.0)$$

$$= 2.032 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.09)1.0$$

$$= 7.857 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.0258 mm.
Wall Thickness per UG16(b),	tr16b = 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.2373 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.2373 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3503 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm.
Wall Thickness per table UG-45	tb3 = 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.4, \max(8.2373, 4.1)]$$

$$= 6.4000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.0258, 6.4)$$

$$= 6.4000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 12.0, tr = 6.637, c = 1.6 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.638, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 4.57/51.00 = 0.090$$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 10.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (3.5841 - 2.0319 + 2 \cdot 15.2 \cdot 1.0 \cdot (1.0 \cdot 10.4 - 6.6373)) \cdot 138)$$

$$= 3793.88 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (7.8575 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 12463.84 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$



$$= (7.8575 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (7.8575 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16912.71 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 3793 Kgf	or W1 = 12463 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 3793 Kgf	or W2 = 16912 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 3793 Kgf	or W3 = 16912 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.0932 mm.

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Input, Nozzle Desc: K1B-2"

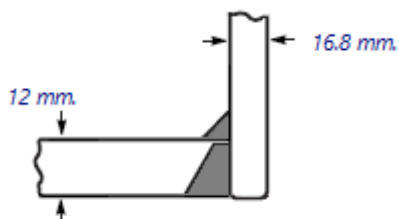
From: 50

Pressure for Reinforcement Calculations	P	4.925	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4000.00	mm.
Design Length of Section	L	1548.3328	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		9450.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B-2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4.93 \cdot 2001.6001 / (138 \cdot 1.0 - 0.6 \cdot 4.93)$
= 7.1597 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
= $27.0 \left(\exp \left(\frac{4.93}{138.0 \cdot 1.0} \right) - 1 \right)$
= 0.0965 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2975 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.0000 mm.
Parallel to Vessel Wall, opening length	d	54.0000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.0000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

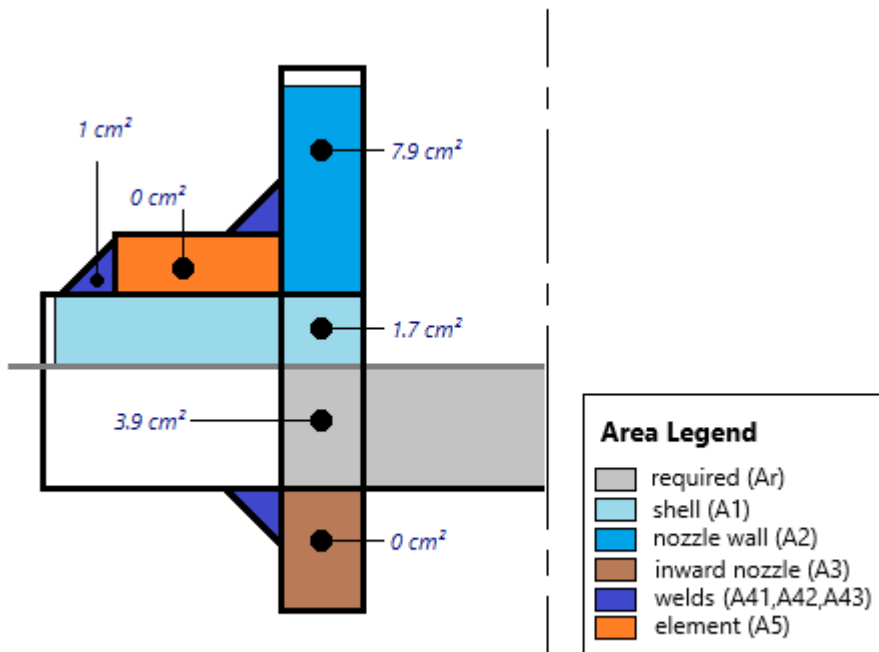


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3.866	1.784	NA
Area in Shell	A1	1.750	2.047	NA
Area in Nozzle Wall	A2	7.854	7.749	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.604	10.797	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (54.0 \cdot 7.1597 \cdot 1.0 + 2 \cdot 15.2 \cdot 7.1597 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 3.866 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 54.0(1.0 \cdot 10.4 - 1.0 \cdot 7.16) - 2 \cdot 15.2$$

$$(1.0 \cdot 10.4 - 1.0 \cdot 7.1597) \cdot (1 - 1.0)$$

$$= 1.750 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.0)(15.2 - 0.1)1.0$$

$$= 7.854 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$



= 1.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.8975$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.7597$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.7597$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 2.3503$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.1000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb_3 , max(tb_1 , tb_2)]
 = min[6.4, max(8.7597, 4.1)]
 = 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(t_a , t_b)
 = max(1.8975, 6.4)
 = 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 12.0$, $t_r = 7.16$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.688$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $4.93/51.00 = 0.097$

Weld Size Calculations, Description: K1B-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.4000$ mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, ($A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)$) S_v)
 = max(0, ($3.8662 - 1.7498 + 2 \cdot 15.2 \cdot 1.0 \cdot$
 ($1.0 \cdot 10.4 - 7.1597$)) 138)
 = 4364.35 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= ($A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2$) $\cdot S_v$
 = ($7.8538 + 0.0 + 1. - 0.0 \cdot 1.0$) $\cdot 138$
 = 12458.68 Kgf

Weld Load [W2]:

= ($A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)$) $\cdot S_v$



$$= (7.8538 + 0.0 + 1. + (3.1616)) \cdot 138$$

$$= 16907.55 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S$$

$$= (7.8538 + 0.0 + 1. + 0.0 + (3.1616)) \cdot 138$$

$$= 16907.55 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 9141. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 138$$

$$= 16275. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 14357. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9141 + 16275) = 25416 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9141 + 0 + 14357 + 0) = 23499 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9141 + 14357 + 0) = 23499 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 25416	Kgf, must exceed W = 4364 Kgf	or W1 = 12458 Kgf
Path 2-2 = 23498	Kgf, must exceed W = 4364 Kgf	or W2 = 16907 Kgf
Path 3-3 = 23498	Kgf, must exceed W = 4364 Kgf	or W3 = 16907 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 0.4451 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.8202 mm.

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Input, Nozzle Desc: M2-24"

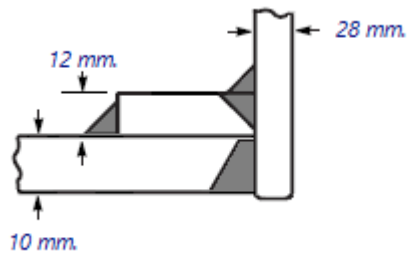
From: 70

Pressure for Reinforcement Calculations	P	4.731	bars
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.52	bars
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	4000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.73 \cdot 4003.2 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.73)$
 $= 6.1742 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4.73 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 4.73)$
 $= 1.0437 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0288 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm.
Parallel to Vessel Wall, opening length	d	556.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		21.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

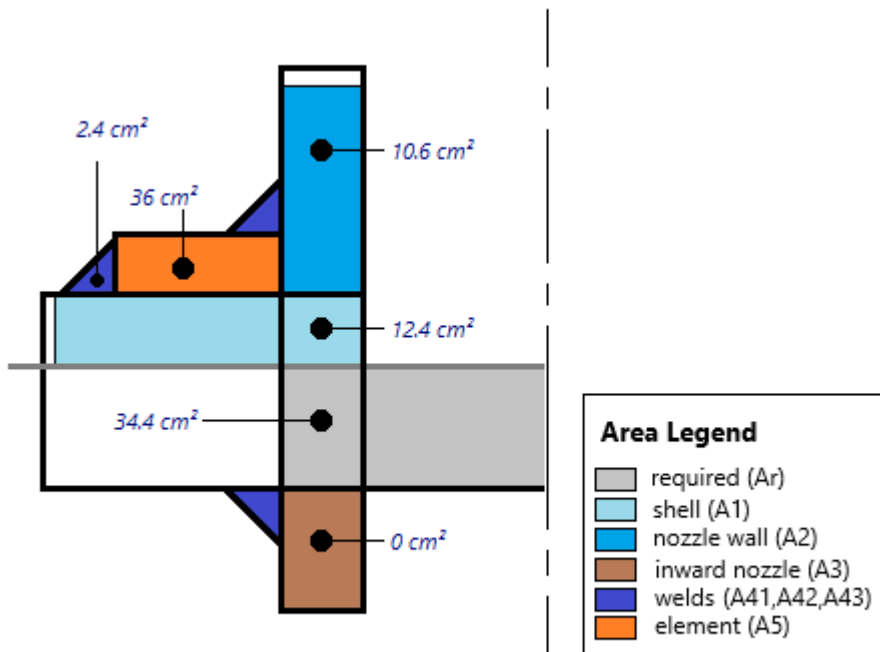


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	34.378	20.490	NA
Area in Shell	A1	12.393	5.791	NA
Area in Nozzle Wall	A2	10.650	10.656	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.430	2.430	NA
Area in Element	A5	36.000	36.000	NA
TOTAL AREA AVAILABLE	Atot	61.473	54.877	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	683.8120	12.0000 mm.
Based on given Pad Diameter:	909.6000	2.9685 mm.
Based on Shell or Nozzle Thickness:	698.6544	10.0000 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (556.8 \cdot 1.0 \cdot 8.4 - 1.0 \cdot 2 \cdot 26.4 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 34.378 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 556.8(1.0 \cdot 8.4 - 1.0 \cdot 6.174) - 2 \cdot 26.4$$

$$(1.0 \cdot 8.4 - 1.0 \cdot 6.1742) \cdot (1 - 1.0)$$

$$= 12.393 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 21.0) \cdot (26.4 - 1.04) \cdot 1.0$$

$$= 10.650 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr}_3 + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr}_2 + W_p^2 \cdot \text{fr}_4$$

$$= (0.99) \cdot 1.0 + (0.0) \cdot 1.0 + 12.0^2 \cdot 1.0$$

$$= 2.430 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr}_4$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 36.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.6437 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.6437 \text{ mm}$$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 28.0$, $t_r = 1.044$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.04$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 12.0$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.174$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:D, Curve:B)

Govrn. thk, $t_g = 10.0$, $t_r = 6.174$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.735$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4.73/19.60 = 0.241$

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 10.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$5.2000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (34.3782 - 12.393 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 8.4 - 6.1742)) \cdot 138)$
 $= 32590.19 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (10.6497 + 36.0 + 2.43 - 0.0 \cdot 1.0) \cdot 138$
 $= 69062.80 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (10.6497 + 0.0 + 1. + (4.4352)) \cdot 138$
 $= 22633.93 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (10.6497 + 0.0 + 2.43 + 36.0 + (4.4352)) \cdot 138$
 $= 75303.81 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 909.6 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 118220. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 238226. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 119654. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (10.0 - 1.6) \cdot 0.74 \cdot 138$
 $= 83758. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (118220 + 238226) = 356446 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (66025 + 119654 + 83758 + 0) = 269438 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (118220 + 83758 + 0) = 201978 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 356446 Kgf, must exceed $W = 32590 \text{ Kgf}$ or $W1 = 69062 \text{ Kgf}$

Path 2-2 = 269437 Kgf, must exceed $W = 32590 \text{ Kgf}$ or $W2 = 22633 \text{ Kgf}$

Path 3-3 = 201977 Kgf, must exceed $W = 32590 \text{ Kgf}$ or $W3 = 75303 \text{ Kgf}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 12.8977 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 272.8978 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.814 %

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**PV Elite[®] Pressure Vessel Design Notes**

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: